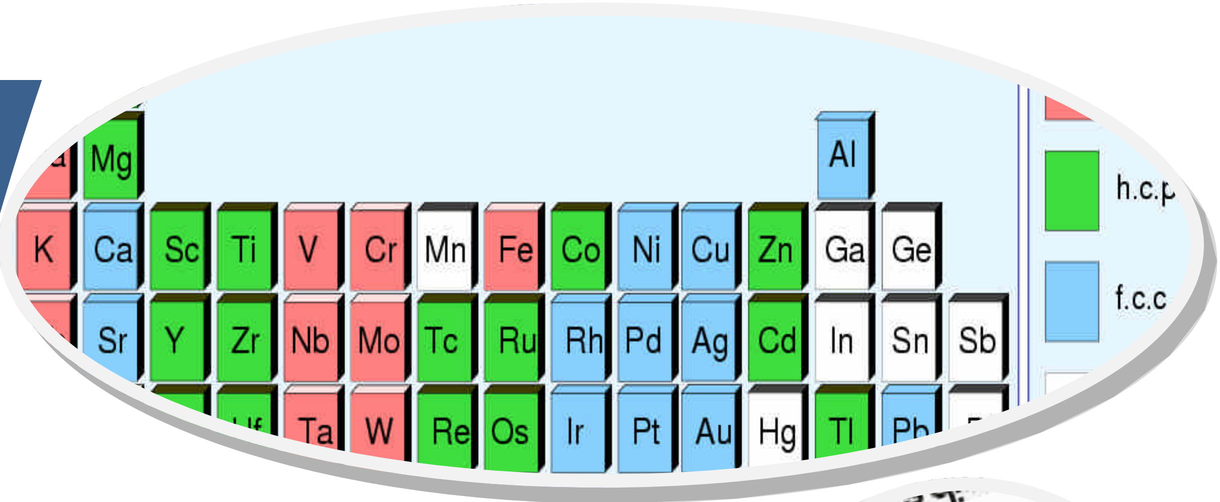
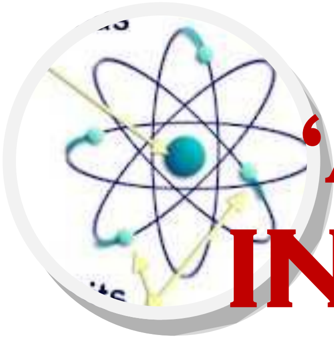
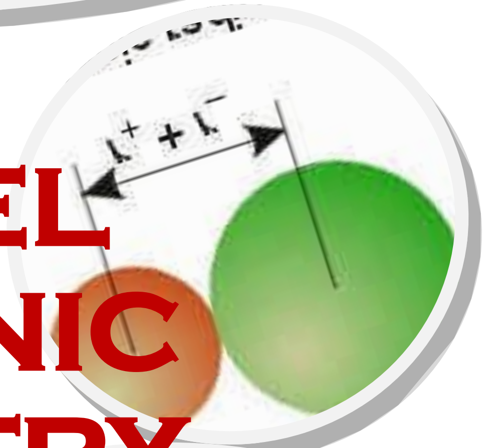
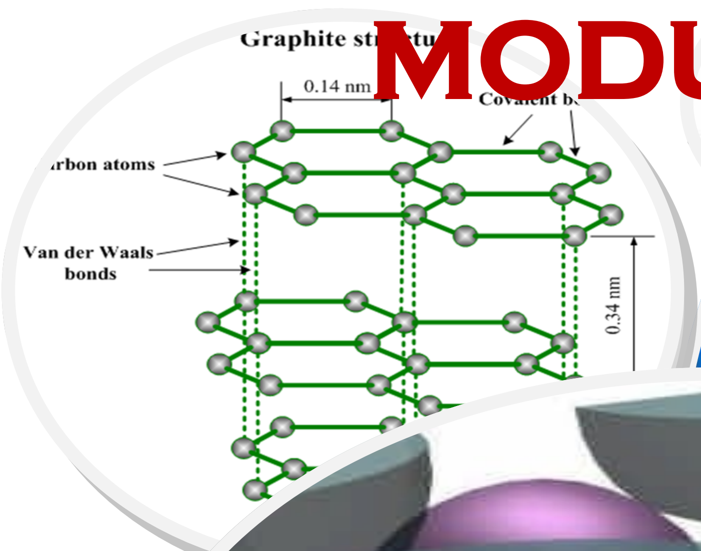




# 'A' LEVEL INORGANIC CHEMISTRY MODULE




BY MUKANDA S.M

# INORGANIC CHEMISTRY

By

MUKANDA S. M

lsmichael2005@yahoo.co.uk

cell: +263778494504

# Aims and Intentions

Most General Advanced level texts do not address all the objectives spelt in the syllabus especially in inorganic chemistry therefore it requires the consultation of a number of dedicated Inorganic Chemistry textbooks to cover all the objectives, most of which are unavailable in High School libraries and are also quite reader unfriendly.

This booklet addresses the underlying principles of Advanced level Inorganic Chemistry as it looks at the fundamental aspects determining trends in physical and chemical properties as well as relevance of the elements and compounds of the main Groups to our day to day lives. Meticulous effort was made to address every Inorganic chemistry aspect mentioned in the Advanced level Chemistry syllabus

***By***
***MUKANDA S.M***
***(BSc Hons. Chemistry)***
***e-mail:*** lsmichael2005@yahoo.co.uk
***cell:*** **+263778494504**

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# INORGANIC CHEMISTRY

It is the study of elements and their compounds other than organic compounds focusing mainly on the periodic table.

**Periodic Table: Brief History**

Chemists have since a long time ago been looking for ways of dividing up elements into groups. The first idea was to group them into metals and non-metals but other elements were discovered which had properties between metallic and non-metallic properties.

One of the earliest classifications was put forward by J. Newlands a British chemist in 1865. He arranged the known elements in order of increasing atomic weight and showed that similar physical and chemical properties were found in every eight elements in the series. This led him to propose the '**LAW OF OCTAVES'** but his suggestion was received with ridicule *(an octave is a set of musical 8 notes apart).* It was the Russian scientist, Dimitri Mendeleev who in 1869 developed Newland's idea and presented it with such vigour and confidence that it soon won universal acceptance. It is upon Mendeleev's Periodic Table that the present day Periodic Table is built as the undiscovered elements were fitted into the gaps he had left out. He summarised his proposal as "*the properties of the elements are a periodic function of their atomic weights."*

The present day Periodic Table consists of 116 known elements, 92 of which are naturally occurring while the rest are man-made.

**Present day Periodic Table**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 H 1·008 | | | | | | | | | | | | | | | | | 2 He 4·003 |
| 3 Li 6 | 4 Be 9·012 | | | | | | | | | | | 5 B 10·81 | 6 C 12·01 | 7 N 14·01 | 8 O 16·00 | 9 F 19·00 | 10 Ne 20·18 |
| 11 Na 22·99 | 12 Mg 24·31 | | | | | | | | | | | 13 Al 26·98 | 14 Si 28·09 | 15 P 30·97 | 16 S 32·06 | 17 Cl 35·45 | 18 Ar 39·94 |
| 19 K 39·10 | 20 Ca 40·08 | 21 Sc 44·96 | 22 Ti 47·90 | 23 V 50·94 | 24 Cr 52·00 | 25 Mn 54·94 | 26 Fe 55·85 | 27 Co 58 93 | 28 Ni 58·71 | 29 Cu 63·55 | 30 Zn 65·38 | 31 Ga 69·72 | 32 Ge 72·59 | 33 As 74·92 | 34 Se 78·96 | 35 Br 79·90 | 36 Kr 83·80 |
| 37 Rb 85·47 | 38 Sr 87·62 | 39 Y 88·91 | 40 Zr 91·22 | 41 Nb 92·91 | 42 Mo 95·94 | 43 Tc (97) | 44 Ru 101·1 | 45 Rh 102·9 | 46 Pd 106·4 | 47 Ag 107·9 | 48 Cd 112·41 | 49 In 114·8 | 50 Sn 118·7 | 51 Sb 121·8 | 52 Te 127·6 | 53 I 126·9 | 54 Xe 131·3 |
| 55 Cs 132·9 | 56 Ba 137·3 | 57 La 138·9 | 72 Hf 178·5 | 73 Ta 180·5 | 74 W 183·9 | 75 Re 186·2 | 76 Os 190·2 | 77 Ir 192·22 | 78 Pt 195·1 | 79 Au 197·0 | 80 Hg 200·6 | 81 Tl 204·4 | 82 Pb 207·2 | 83 Bi 209·0 | 84 Po (209) | 85 At (210) | 86 Rn (222) |
| 87 Fr (223) | 88 Ra 226·0 | 89 Ac 227·03 | | | | | | | | | | | | | | | |
| | | | | 58 Ce 140·1 | 59 Pr 140·9 | 60 Nd 144·2 | 61 Pm 145 | 62 Sm 150·4 | 63 Eu 152·0 | 64 Gd 157·3 | 65 Tb 158·9 | 66 Dy 162·5 | 67 Ho 164·9 | 68 Er 167·2 | 69 Tm 168·9 | 70 Yb 173·0 | 71 Lu 175·0 |
| | | | | 90 Th 232·0 | 91 Pa 231·0 | 92 U 238·0 | 93 Np 237·0 | 94 Pu (244) | 95 Am (243) | 96 Cm (247) | 97 Bk (247) | 98 Cf (251) | 99 Es (254) | 100 Fm (257) | 101 Md (258) | 102 No (259 | 103 Lr (260) |

## Importance of Periodic Table

- It provides a logical frame work for recognising patterns in properties of elements and their compounds.
- It allows us to explain trends and similarities in properties of elements and their compounds in terms of the electronic configuration of the elements
- It helps to tie together concepts in and unite chemistry organisation of the Periodic Table
- It shows the electronic structure of each element.

## STRUCTURE OF PERIODIC TABLE

In the Periodic Table the elements are arranged in horizontal rows called **'Periods'** (*atoms of elements having the same number of shells*) and vertical columns called **'Groups'** (*atoms of elements having the same electronic configuration*).

## Blocks in the Periodic Table

- ***Reactive Metals***

  The elements in Group I (*alkali metals*) and II (*alkaline earth metals*) makes up the reactive metals block. They are also referred to as the **'s-block'** elements, since the outermost electrons in these metals are in **'s'** sub-shells. These metals form stable involatile ionic compounds and are all high up in the electrochemical series. They have lower melting and boiling points as well as their densities.

- ***Transition Metals***

These elements are found between Group II and III in the Periodic Table. They are also called the **'d-block'** because electrons are being added into the **'d'** sub-shells. The elements also have the ability to exhibit different oxidation states as they can use different number of electrons in chemical reactions.

- ***Poor Metals***

These are found to the right of the Transition metals and are called 'poor metals' as they exhibit strong metallic characteristics, relatively unreactive and many of their reactions resemble those of non-metals. They are usually known as **'p-block'** as their outer electrons are added into their **'p'** sub-shells. They are found at the bottom of the electrochemical series as they are less reactive.

- ***Metalloids***

These are diagonally found in the Periodic Table between Poor Metals and Non-metals. They have chemical properties midway between those of metals and non-metals, includes: Boron, Carbon (*graphite*), Silicon, Germanium, Arsenic and Antimony. Their metallic nature increases with temperature e.g conductivity and they are also known as **'p-block'** because outer shell electrons are added into the **'p'** sub-shells.

- ***Non-Metals***

These elements form a triangular section to the right in the Periodic Table. The elements are called **'p block'** as the outermost electrons in these elements are going into the **'p'** sub-shells.

- ***Noble Gases***

The atoms of these elements have completely filled **'s'** and **'p'** sub-shells of electrons. They are extremely unreactive as a result of their completely filled sub-shells.

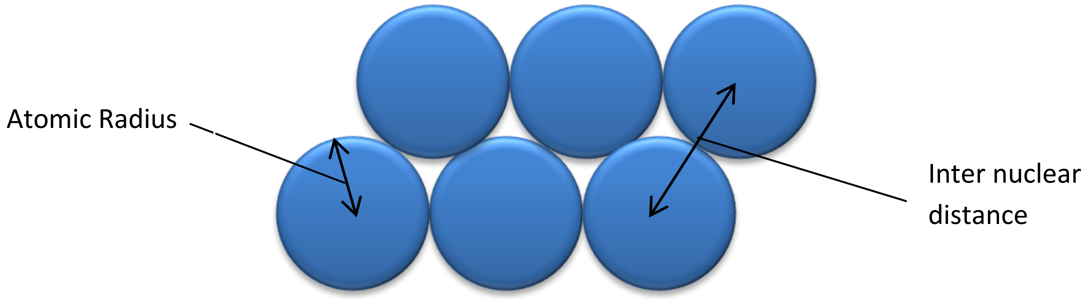
## TRENDS ACROSS THE 3RD PERIOD Na-Ar

| ELEMENTS | Na | Mg | Al | Si | P | S | Cl | Ar |
|---|---|---|---|---|---|---|---|---|
| TYPE OF ELEMENT | Metals | | | Metalloid | Non-Metal | | | |
| STRUCTURE AND BONDING | Giant Metallic | | | Giant Molecular | Simple Molecular<br>$P_4$ | $S_8$ | $Cl_2$ | Ar |
| STATE @ R.T.P | Solids | | | | | | Gases | |
| MELTING POINT /°C | 98 | 650 | 660 | 1410 | 44 | 119 | -101 | -189 |
| BOILING POINTS /°C | 890 | 1120 | 2450 | 2680 | 280 | 445 | -34 | -186 |
| ELECTRICAL CONDUCTIVITY X $10^3$ $ohm^{-1}$ $cm^{-4}$ | 10 | 16 | 38 | 0.4 | $10^{-16}$ | $10^{-22}$ | - | - |
| ATOMIC RADIUS /nm | 0.186 | 0.16 | 0.143 | 0.117 | 0.110 | 0.104 | 0.099 | 0.192 |
| IONIC RADIUS /nm | 0.095 | 0.065 | 0.050 | 0.041 | 0.212 | 0.184 | 0.181 | - |
| 1ST I.E (KJ/Mol) | 494 | 736 | 577 | 786 | 1060 | 1000 | 1260 | 1520 |

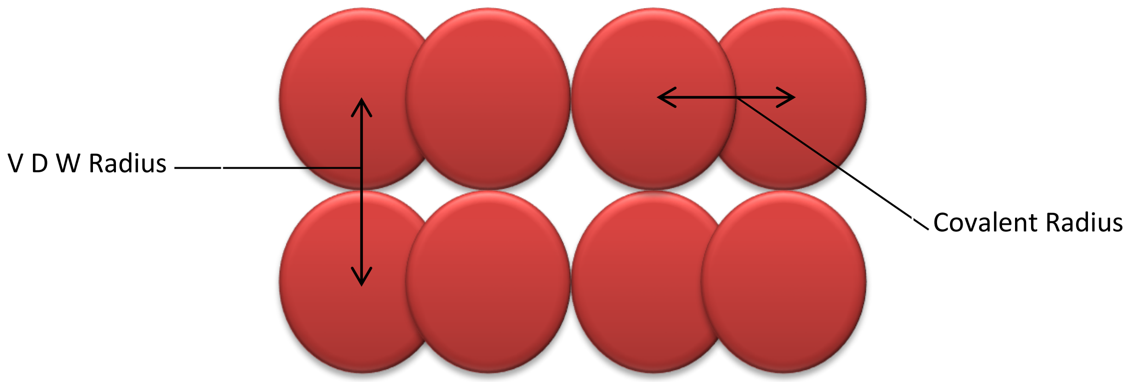
## PERIODICITY OF PHYSICAL PROPERTIES OF THE ELEMENTS

- ***Variation in Atomic Radii***

  Atomic radii of atoms are obtained by X-ray analysis and electron density maps.
  ***Def:*** *Atomic radius of an atom is the distance of closest approach by another identical atom*
  The atomic radii of metals are obtained by measuring the distance between the nuclei of neighbouring atoms in the metal crystals i.e the atomic radius is simply half of the internuclear distance(*Metallic Radii* )




The atomic radii of non-metals are obtained from the distance between the nuclei of similar atoms joined by a covalent bond i.e the atomic radius is half of the covalent bond length (*Covalent Radii)*. The distance in non-metal crystals is called the *Van der Waals* radius (VDW Radii- is *half the distance between the nuclei of two neighbouring similar atoms which are not chemically bonded)*




From Na to Cl atomic radius decreases gradually and then increases significantly from Cl to Ar due to the increase in **Effective Nuclear Charge.**

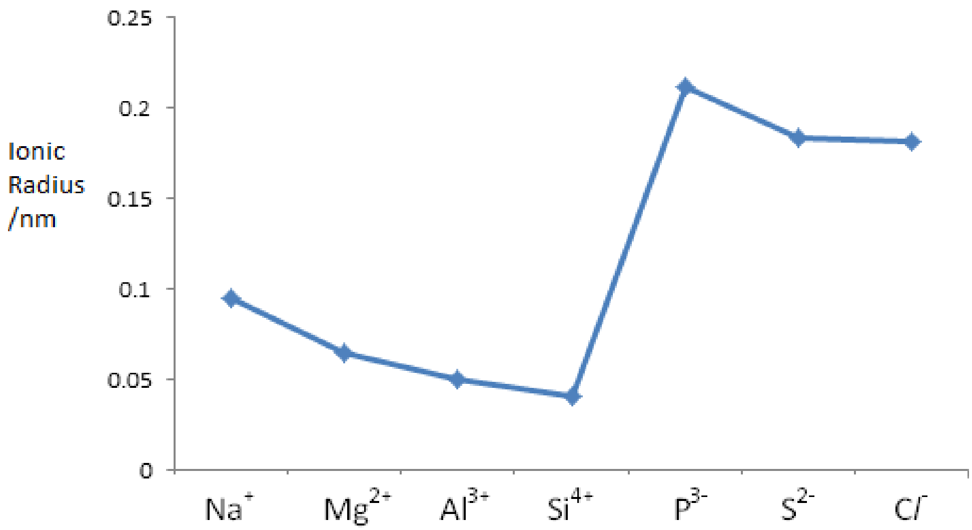
**Effective Nuclear Charge-** is the actual nuclear force falling upon a particular electron.

- Moving across the period electrons are being added into the same shell thus shielding effect remains almost constant compared to the increasing nuclear charge due to increase in the proton number hence nuclear charge outweighs the shielding or screening effect thus result in a decrease in atomic radii.
- The increase from Cl to Ar is due to the fact that we are now referring to Van der Waals radius. Van der Waals interactions are weaker than covalent bonds hence the long inter-nuclear radius.

★ **NB** :- *Down the group atomic radius increases as outer electrons enter new energy levels thus screening increases greatly although proton number also increases down the group.*

❖ ***Variation in Ionic Radius***

Variation in ionic radii is best illustrated with a sketch shown below.




- In general the ionic radii of the cations are smaller than those of the corresponding atom. This is because loss of electrons results in an increased effective nuclear charge as there is decreased screening effect hence increases the force of electrons on the outer shell electrons causing a decrease in ionic radii.
- For the anions the radius of the anion is larger than that of the corresponding atom. In this case the effective nuclear charge decreases due to increased screening effect resulting from the gained electrons. As the number of the electrons gained decreases so does the ionic radius of the anion. A corresponding increase in the nuclear charge also adds to this effect.

❖ ***Variation of the 1st I.E***

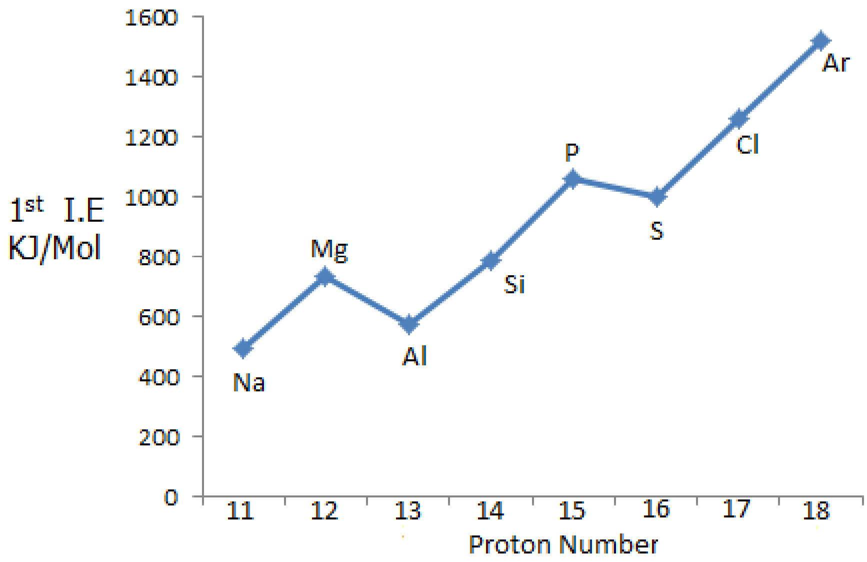
**First I.E ($\Delta H_{IE}$):-** is the minimum amount of energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous univalent cations.

E.g. $Na_{(g)} \longrightarrow Na^{+}_{(g)} + e^{-}$

The magnitude of 1st I.E depends on the following:-

- Nuclear charge
- Screening effect
- Electronic Configuration

The trend in the 1st I.E for Period 3 elements are shown below:




- Generally going across the period there is a general increase in the first I.E due to increase in the effective nuclear charge. The increase in the nuclear charge outweighs the screening effect since electrons being added into the same shell. Electrons in the same shell have minimum screening effect on each other.

- *Discontinuity in the trends between Mg and Al*

  This can be explained in terms of the electronic configuration of these elements.

  $_{12}Mg$ – $1s^2\ 2s^2\ 2p^6\ 3s^2$
  $_{13}Al$ - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
  The electron to be removed in Mg is from a lower energy fully filled $3s^2$ which is more stable than the higher energy partially filled $3p^1$ orbital of Al

- *Discontinuity in the trends between P and S*

  Electronic configuration of P and S

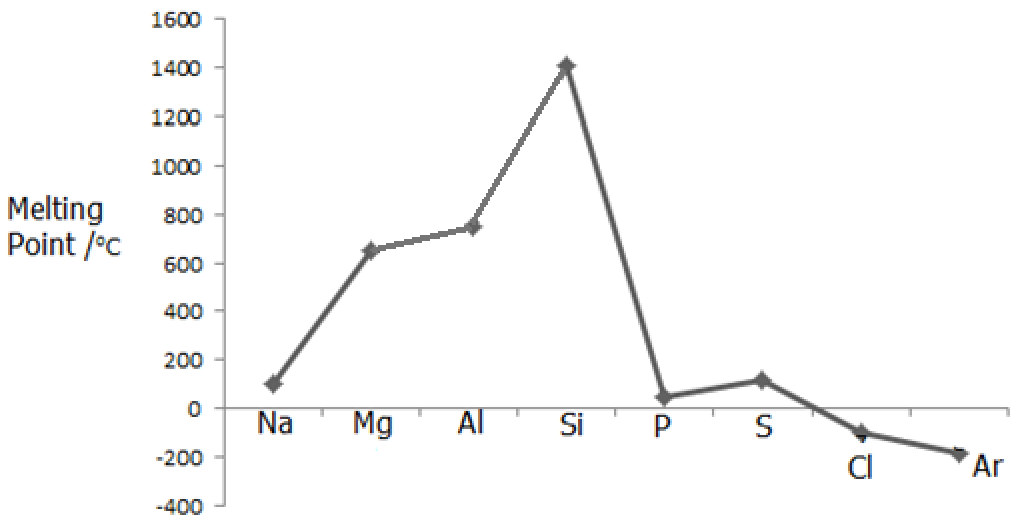
  $_{15}P$ - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
  $_{16}S$ - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
  The electron to be removed in Sulphur is paired in the 3p orbital hence there is inter-electron repulsions making the removal easier thus Sulphur has a lower I.E than in Phosphorus where the electron to be removed in Phosphorus is from a half filled $3p^3$ orbital (*Half-filled orbital have a more symmetrical charge distribution than partially filled hence more stable*) with greater stability hence a larger 1st I.E.

**NB:-** *1st I.E decreases down the group due to increase in the atomic radii as a result of an increase in the number of electron shells thus effective nuclear charge decreases. Shielding effect increases hence outer electrons are less effectively attracted to the positive nucleus hence 1st I.E is lower.*

## ❖ *Variation in Melting Point*

- Melting is a physical process which occurs when a substance changes state from solid to liquid. To melt a substance inter-particle forces in the solid lattice must be broken, be they hydrogen bonds, electrostatic force, covalent bonds or Van der Waals forces. The amount of heat required for this change is thus related to the type of forces holding particles together. The melting and boiling point of a substance is best explained in terms of the type of structure and bonding present.

*Graph of variation in melting point*



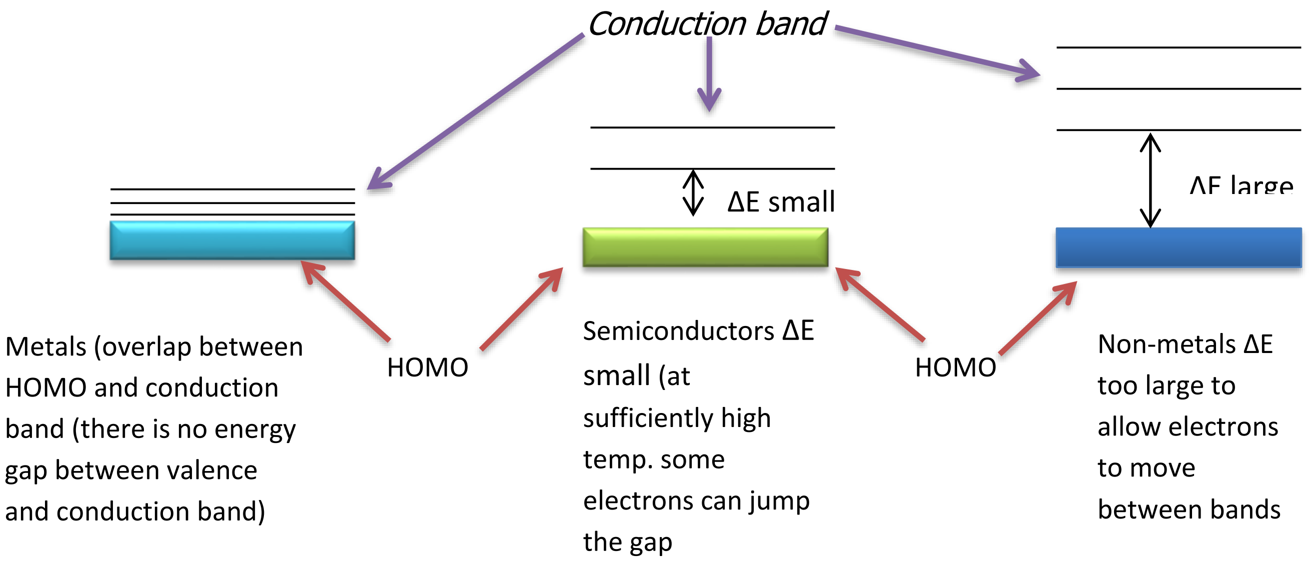

- Na, Mg and Al being giant metallic structure have fairly high melting points due to strong metallic bonding holding the metal atoms together. The strength of metallic bonding increases from Na to Al as number of delocalised valence electrons in the metallic structure increases.
- Silicon has the highest melting point in the period due to the strong network of covalent bonds holding the Si atoms in the giant molecular structure.
- Phosphorus, Sulphur, Chlorine and Argon are simple molecular structures which consist of discrete molecules held together by weak Van der Waals forces hence their low melting points. The strength of the Van der Waals forces varies depending on the number of electrons in a molecule. Phosphorus exists as $P_4$ molecules (*with 60 electrons*), Sulphur as $S_8$ (*with 128 electrons having the highest MP*), Chlorine as $Cl_2$ (*with 34 electrons*), and Argon as monoatomic Ar molecules (*with 18 electrons*) has the lowest melting point.

## ❖ *Variation of Electrical Conductivity*

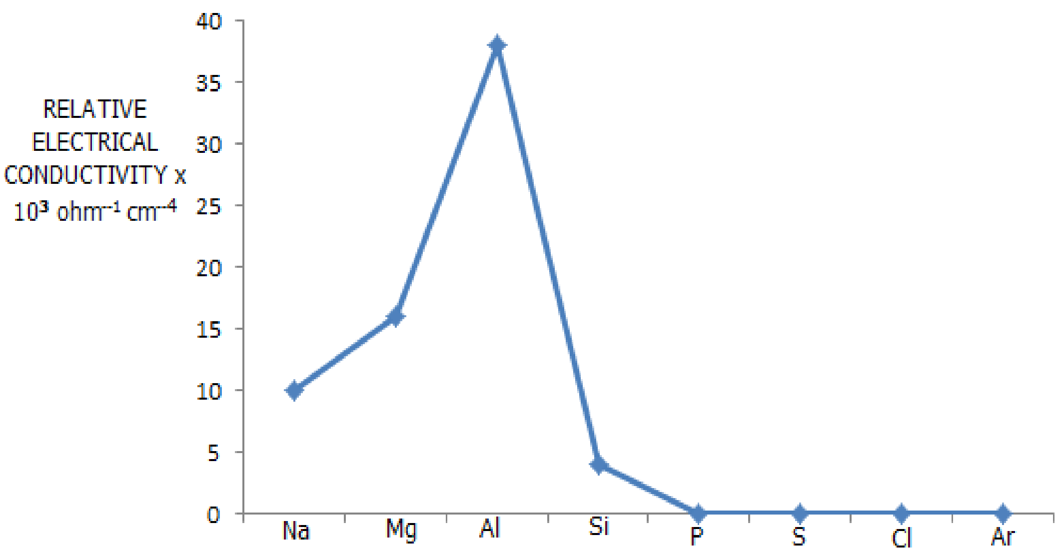
- The metals Sodium, Magnesium and Aluminium being giant metallic structures are very good electrical conductors due to presence of delocalised valence electrons in the metallic lattice.

The electrical conductivity increases gradually from Na to Al due to increase in the number of delocalised electrons.

- Silicon is a semiconductor with conductivity less than that of metals but higher than that of non-metals. The conductivity arises from promotion of electrons in the highest occupied molecular orbital (HOMO) into the conduction bands due to the small energy gap between these energy levels.
- Phosphorus, sulphur, Chlorine and Argon are virtual non-conductors due to absence of delocalised electrons. The energy gap between the highest occupied molecular orbital and conduction band is too large.




*Trend graph in Electrical conductivity of Period 3 Elements*




❖ ***Electronegativity***

**Def:-** this is a measure of the tendency of an atom to attract electrons in a covalent bond. The larger the positive charge on the nucleus the more effectively will it attract electrons.

- Electronegativity increases across a period due to increase in effective nuclear charge as atomic radius decreases with increase in proton number.

| Element | Na | Mg | Al | Si | P | S | Cl | Ar |
|---|---|---|---|---|---|---|---|---|
| Electronegativity | 1.0 | 1.3 | 1.5 | 1.7 | 2.0 | 2.5 | 3.0 | -- |

★ **NB:-** *Electronegativity decreases down the group due to increase in atomic radius resulting in a decrease in effective nuclear charge hence elements down the group tends to loose electrons more readily as screening effect increases.*

## CHEMICAL PERIODICITY OF PERIOD 3 ELEMMENTS

❖ ***Reaction of the elements with Oxygen***

The elements Na to S all reacts with oxygen (except Cl and Ar).

- **Sodium:-** reacts very vigorously forming $Na_2O$. To prevent its oxidation it is usually stored under oil or paraffin. When heated in air it burns with a brilliant yellow flame forming a mixture of oxide and peroxides of which the peroxide predominates.

$$2Na_{(s)} + \tfrac{1}{2}O_{2(g)} \longrightarrow Na_2O_{(s)}$$
$$2Na_{(s)} + O_{2(g)} \longrightarrow Na_2O_{2(s)}$$

- **Magnesium:-** is also very reactive towards oxygen but its reactivity is less than Na. It ignites with a brilliant white dazzling flame when heated above its melting point forming a white powder of MgO.

$$Mg_{(s)} + \tfrac{1}{2}O_{2(g)} \longrightarrow MgO_{(s)}$$

- **Aluminium:-** Although Al is highly electropositive its reactivity is hindered by the formation of an impervious oxide layer on its surface which protects the metal atoms in the interior from further corrosion. It can however be converted to the oxide by heating up to 800°C.

$$4Al_{(s)} + 3O_{2(g)} \longrightarrow 2Al_2O_{3(s)}$$

- **Silicon:-** reacts slowly with oxygen on heating giving $SiO_2$.

$$Si_{(s)} + O_{2(g)} \longrightarrow SiO_{2(s)}$$

- **Phosphorus:-** burns in air forming a solid which easily sublimes to form dense white smoke. In limited oxygen it forms $P_4O_6$ and in excess oxygen it forms $P_4O_{10}$.

$$4P_{(s)} + 3O_{2(g)} \longrightarrow P_4O_{6(s)}$$
$$4P_{(s)} + 5O_{2(g)} \longrightarrow P_4O_{10(s)}$$

- **Sulphur:-** burn in air with a blue flame forming $SO_2$ and a little $SO_3$

$$S_{(s)} + O_{2(g)} \longrightarrow SO_{2(g)}$$

$$2S_{(s)} + 3O_{2(g)} \longrightarrow 2SO_{3(g)}$$

Oxidation numbers of elements in their oxides are always positive and correspond to the number of electrons used for bonding. The maximum oxidation number increases across the period from +1 to 16.

Oxygen is highly electronegative hence tend to draw electrons to itself thus leaving the atoms of the elements partially positive.

★ **NB:** *Phosphorus and Sulphur show a wide range of oxidation states in their oxides and chlorides because they are able to expand their octet by utilizing the vacant low lying 3d orbitals in the 3rd shell which are available for bonding in certain circumstances forming more covalent bonds.*

❖ ***Reaction of the elements with $H_2O$***

- **Sodium:** reacts vigorously with cold water liberating hydrogen gas. The sodium metal moves about the water surface due to rapid evolution of hydrogen gas and heat evolved by reaction. The resulting solution is strongly basic/alkaline.

$$2Na_{(s)} + 2H_2O_{(l)} \longrightarrow 2NaOH_{(aq)} + H_{2(g)} \quad pH \approx 13.4$$

- **Magnesium:** reacts very slowly with cold water giving a weakly alkaline solution of $pH \approx 8$.

$$Mg_{(s)} + 2H_2O_{(l)} \longrightarrow Mg(OH)_{2(aq)} + H_{2(g)}$$

In steam the reaction occurs appreciably fast but giving MgO and $H_{2(g)}$

$$Mg_{(s)} + H_2O_{(l)} \longrightarrow MgO_{(s)} + H_{2(g)}$$

- **Aluminium:-**is affected in reactivity towards water or steam by the formation of the impervious oxide layer on its surface.
- **Silicon, phosphorus and sulphur** do not react with water. However $Cl_2$ dissolves in $H_2O$ forming a mixture of hydrochloric acid and hypochloric acid.

$$Cl_{2(g)} + H_2O_{(l)} \Longleftrightarrow HClO_{(aq)} + HCl_{(aq)}$$

**PROPERTIES OF OXIDES OF ELEMENTS IN PERIOD 3**

| OXIDE | $Na_2O$ | $MgO$ | $Al_2O_3$ | $SiO_2$ | $P_4O_{10}$ ($P_4O_6$) | $SO_2$ ($SO_3$) |
|---|---|---|---|---|---|---|
| **Ox. No.** | +1 | +2 | +3 | +4 | +5/+3 | +2/+3 |
| **State @ r.t.p** | solid | solid | solid | solid | Solid (both) | gas (liquid) |
| **Structure of oxide** | Giant ionic | | | Giant Molecular | Simple Molecular | |
| **Melting Point/°C** | 1132 | 2900 | 2080 | 1700 | 58 (24) | -75 (17) |
| **Boiling Point/°C** | Decomposes | 3600 | 2980 | 2230 | sublimes | -10 (75) |
| **Acid-base Nature** | Basic (alkaline) | | Amphoteric | Acidic | | |
| **Electrical conductivity of oxide in liquid state** | Good | | | Very Poor | Nil | Nil |
| **Effect of adding oxide to water** | React to form $NaOH_{(aq)}$ alkaline solution of pH≈13 | Slightly dissolves to give a weakly alkaline solution of $Mg(OH)_{2(aq)}$ pH≈9 | Insoluble pH=7 | | Readily react with water to form strongly acidic solutions of pH≈2 | |

❖ ***Variation in Melting Points of the oxides***

- $Na_2O$, $MgO$ and $Al_2O_3$ have giant ionic structures in which cations form electrostatic forces to the $O^{2-}$ ion. Due to the strength of the electrostatic forces they have relatively high melting point. MgO has highest melting point of the 3 oxides due to the high charge density of the $Mg^{2+}$ ion. For $Al^{3+}$, the bonding is ionic with a significant covalent character resulting from the high polarity power of $Al^{3+}$. The covalent character weakens the electrostatic force hence its lower melting point.
- $SiO_2$ also has a relative high melting point (1700°C) this resulting from a lot of energy being required to break the network of Si—O bonds in the giant molecular structure similar to that of diamond.

- The oxides of Phosphorus and Sulphur are all simple molecular with low melting point due to weak Van der Waal's forces holding the molecules together. The strength of the Van der Waals' forces increases with increase in number of electrons in the molecule thus oxides of phosphorus has a higher melting point than oxides of sulphur.

## ❖ ***Reaction of oxides with Water***

- $Na_2O$ being ionic dissolve in water to form a strongly alkaline solution.

$$Na_2O_{(s)} + H_2O_{(l)} \longrightarrow 2NaOH_{(aq)} \quad pH \approx 13$$

- MgO is only slightly soluble (sparingly) giving a weakly basic solution.

$$MgO_{(s)} + H_2O_{(l)} \rightleftharpoons Mg(OH)_{2(aq)} \quad pH \approx 8$$

- $Al_2O_3$ is insoluble in water due to a large lattice enthalpy change as a result of a high charge density.
- $SiO_2$ has a giant molecular structure and $H_2O$ molecules are not able to break (hydrolyse) the network of Si—O—Si bonds in the giant molecular structure.
- The remaining oxides which are simple molecular are easily hydrolysed by water to give strongly acidic solutions. They hydrolysis reaction occurs through use of accessible low lying vacant 3d-orbital as the oxygen in water molecules donates lone pairs of electrons.

$$P_4O_{6(s)} + 6H_2O_{(l)} \longrightarrow 4H_3PO_{3(aq)} \quad pH \approx 2$$
$$P_4O_{10(s)} + 6H_2O_{(l)} \longrightarrow 4H_3PO_{4(aq)} \quad pH \approx 1$$
$$SO_{2(g)} + H_2O_{(l)} \longrightarrow H_2SO_{3(aq)} \quad pH \approx 2$$
$$SO_{3(g)} + H_2O_{(l)} \longrightarrow H_2SO_{4(aq)} \quad pH \approx 1$$

## ❖ ***Acid-base Nature of the Oxides***

- $Na_2O$ and MgO being ionic oxides are basic and react with acids giving salts and water.

$$Na_2O_{(s)} + 2H^+_{(aq)} \longrightarrow 2Na^+_{(aq)} + H_2O_{(l)}$$
$$MgO_{(s)} + 2H^+_{(aq)} \longrightarrow Mg^{2+}_{(aq)} + H_2O_{(l)}$$

- $Al_2O_3$ is amphoteric and reacts with both $H^+$ and alkalis. Its amphoteric nature results from the mixed type of bonding which is predominantly ionic with a significant covalent character. The covalent character arises from high polarising power of the densely charged $Al^{3+}$ ion which draws electrons strongly from $O^{2-}$ ion introducing a significant covalent character. As a result $Al_2O_3$ behaves as both an ionic and covalent oxide.

(a) *With acids gives a salt and water*

$$Al_2O_{3(s)} + 6H^+_{(aq)} \longrightarrow 2Al^{3+}_{(aq)} + 3H_2O_{(aq)}$$

$Al^{3+}_{(aq)}$ ions in water form an octahedral complex with solvating water molecules. It draws the hydrating water molecules very closely due to its high charge density, forming dative bonds with the water molecules.

$$Al^{3+}_{(aq)} + 6H_2O_{(l)} \longrightarrow [Al(H_2O)_6]^{3+}_{(aq)}$$

(Hexa aqua aluminium III ion)

(b) *With alkalis gives aluminate salt*

$$Al_2O_{3(s)} + 2OH^-_{(aq)} + 3H_2O_{(l)} \longrightarrow 2Al(OH)^-_{4(aq)}$$

- The oxides of Si, P and S being covalent oxides are acidic and react with alkalis forming salts and $H_2O$.

$$SiO_{2(s)} + 2OH^-_{(aq} \longrightarrow SiO_3^{2-}{}_{(aq)} + H_2O_{(l)}$$

$$P_4O_{10(s)} + 12OH^-_{(aq)} \longrightarrow 4PO_4^{3-}{}_{(aq)} + 6H_2O_{(l)}$$

$$P_4O_{6(s)} + 12OH^-_{(aq)} \longrightarrow 4PO_3^{3-}{}_{(aq)} + 6H_2O_{(l)}$$

$$SO_{2(g)} + 2OH^-_{(aq)} \longrightarrow SO_3^{2-}{}_{(aq)} + H_2O_{(l)}$$

$$SO_{3(g)} + 2OH^-_{(aq)} \longrightarrow SO_4^{2-}{}_{(aq)} + H_2O_{(l)}$$

## REACTIONS OF THE ELEMENTS WITH CHLORINE

The elements, Na to P reacts with chlorine to form chlorides. Na and Mg react vigorously forming ionic chlorides. Al, Si and P reacts with chlorine to form covalently bonded molecular chlorides.

### Properties of the Chlorides of elements of Period 3

| CHLORIDE | NaCl | $MgCl_2$ | $Al_2Cl_6$ | $SiCl_4$ | $PCl_3$ ($PCl_5$) |
|---|---|---|---|---|---|
| Ox. No. | +1 | +2 | +3 | +4 | +3 (+5) |
| STATE @ R.T.P | Solid | | | Liquid | Liquid (solid) |
| STRUCTURE | Giant Ionic | | Simple Molecular | | |
| MELTING POINT/°C | 808 | 714 | Sublimes at 180 | -70 | -92 - |
| BOILING POINT/°C | 1465 | 1418 | | 57 | 76 (sublimes @ 164) |
| ELECTRICAL CONDUCTIVITY IN LIQUID STATE | Good | | Very poor | Nil | |
| EFFECT OF ADDING CHLORIDE TO WATER | Readily dissolves giving a neutral solution pH=7 | Readily dissolve with slight hydrolysis to give a slightly acidic solution pH≈6.5 | Reacts with water to give an acidic solution pH≈2 | Completely hydrolysed in water to give a strongly acidic solutions pH≈1 | |

❖ ***Structure and bonding with relation to MP and BP***

- **NaCl and $MgCl_2$:-** are giant ionic structures because of the large difference in the electronegativity between the metallic elements and chlorine hence high melting and boiling points.

- **$Al_2Cl_6$:-** would have been expected to be ionic but in fact it is molecular. This results from the high polarising power of the densely charge $Cl^-$ anion. The $Al^{3+}$ ion draws electrons strongly from the chloride ion introducing a covalent nature in the bonding such that $Al_2Cl_6$ is covalent thus have a simple molecular structure with weak Van der Waals force operating between the molecules hence low MP and BP.

- The remaining chlorides are all simple molecular covalent as the difference in electronegativity between Cl and the elements is very small thus weak Van der Waals forces operate between the molecule hence low MP and BP.

❖ ***Acid-base Nature of the chlorides and their reactions with water***

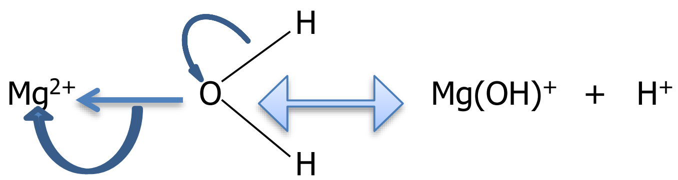
- **NaCl:-** readily dissolve in water without hydrolysis. There is disruption of the ionic lattice forming aquated ion in water.

$$NaCl_{(s)} \quad + (aq) \longrightarrow Na^{+}_{(aq)} \quad + \quad Cl^{-}_{(aq)} \quad pH=7$$

- **$MgCl_2$:-** dissolves readily in water with slight hydrolysis as the Mg— Cl bond is polarised due to the higher charge density of $Mg^{2+}$ ion giving a weakly acidic solution.

$$MgCl_{2(s)} \quad + \quad (aq) \longrightarrow Mg^{2+}_{(aq)} + \quad Cl^{-}_{(aq)} \quad pH\approx 6$$

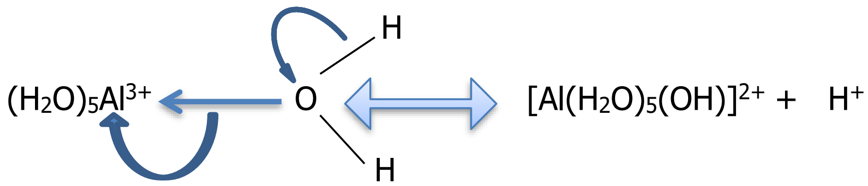
The densely charged $Mg^{2+}$ ion polarises the solvating water molecules causing them to release $H^{+}$ ions.




- **$AlCl_3$:-** is very deliquescent, that is it greatly absorbs moisture. It hydrolyse in water giving strongly acidic solution.

$$AlCl_{3(s)} \quad + \quad 6H_2O_{(l)} \longrightarrow [Al(H_2O)_6]^{3+}_{(aq)} \quad + \quad 3Cl^{-} \quad pH\approx 2$$

Its strong acidity arises again from the high polarising power of densely charged $Al^{3+}$ ion which causes coordinating water molecules to lose $H^{+}$ ions.




- **$SiCl_4$, $PCl_3$/$PCl_5$:-** are easily hydrolyses by water giving strongly acidic solutions. The hydrolysis reactions are facilitated by the availability of low energy vacant 3d orbitals into which water molecules can donate their lone pairs during the hydrolysis reaction.

$$SiCl_{4(l)} \quad + \quad 3H_2O_{(l)} \longrightarrow SiO_2.H_2O_{(s)} \quad + \quad 4HCl_{(aq)} \quad pH\approx 2$$

$$PCl_{3(l)} \quad + \quad 3H_2O_{(l)} \longrightarrow H_3PO_{3(aq)} \quad + \quad 3HCl_{(aq)} \quad pH\approx 2$$

$$PCl_{5(l)} \quad + \quad 4H_2O_{(l)} \longrightarrow H_3PO_{4(aq)} \quad + \quad 5HCl_{(aq)} \quad pH\approx 1$$

## ***USES OF PERIOD 3 ELEMENTS AND THEIR COMPOUNDS***

### Sodium:

1. Liquid sodium used as a coolant in nuclear reactors, manufacture of $Na_4Pb$ which is used to make lead (IV) tetraethyl for leaded petrol.
2. NaOH- used in paper making, purification of bauxite during extraction of Al, manufacture of soaps and bleaches.
3. $Na_2CO_3$- manufacture of glass, NaOH, paints, paper making and as water softener.
4. $NaHCO_3$- used as baking powder and in combating stomach acids
5. NaCl- manufacture of Cl and NaOH, food seasoning and preservation, to melt ice and snow on roads and in glazing earthenware.
6. $Na_3N$- used in car air bags

### Magnesium:

1. Magnesium alloyed with zinc and aluminium used for construction of aircraft and car bodies, used in flare, fireworks and incendiary bombs.
2. $Mg(OH)_2$- used in anti-acid stomach tablets, manufacture of tooth paste
3. $MgSO_4$- used as a laxative, paper filling and anhydrous $MgSO_4$ in treatment of boils.

### Aluminium:

1. Used for making aircraft bodies, making cooking utensils, brewing vats and food processing vessels because compounds of Al are non-poisonous, making electrical conductors and transmission lines, foils used for packaging and capping milk bottles.
2. $Al_2O_3$- source of Al metal, used as an abrasive, refractory material and ceramics, as absorbent in chromatography, manufacture of cement.
3. $Al_2(SO_4)_3$- SEWAGE swage and water purification because it causes coagulation of negatively charged particles, used in leather tanning, water proofing cloth.

### Silicon:

1. Pure Si used to make transistors in electronics.
2. $SiO_2$- manufacture of glass, cement, manufacture of silica-gel used in chromatography. Manufacture of silicones used as lubricants, water repellents and electrical insulators

### Phosphorus:

1. Manufacture of ortho-phosphoric acid ($H_3PO_4$), white phosphorus as rat poison and incendiary bombs.
2. $H_3PO_4$ – main use manufacture of super phosphate fertiliser.

### Sulphur:

1. Burnt directly to form $SO_2$ used in manufacture of sulphuric acid, used in vulcanisation of rubber. In agriculture used as fungicide. Manufacture of safety matches and fireworks.

2.$SO_2$/$SO_3$- manufacture of sulphuric acid, used as food preservative as it prevent fermentation. Sulphuric acid- manufacture of superphosphate and $(NH_4)_2SO_4$ fertiliser, also manufacture of pigments, detergents, drugs, explosives, plastics, lead-acid accumulator as electrolyte.

**Chlorine**:- manufacture of PVC, organic solvents, drugs, antiseptics, bleaches, refrigerants, sterilising drinks, sewage and swimming pool water.

**Argon:-**filling metal filament electric lamps (mixed with $N_2$ to avoid arcing), discharge lamps e.g hallow cathode lamp, for producing inert atmospheres in metallurgy.

# GROUP II- ALKALINE EARTH METALS

This group contains the following elements:

| ELEMENT | Be | Mg | Ca | Sr | Ba | Ra |
|---|---|---|---|---|---|---|
| ELECTRON STRUCTURE | [He] $2s^2$ | [Ne] $3s^2$ | [Ar] $4s^2$ | [Kr] $5s^2$ | [Xe] $6s^2$ | [Rn] $7s^2$ |
| ATOMIC RADIUS/nm | 0.112 | 0.160 | 0.197 | 0.215 | 0.217 | 0.220 |
| IONIC RADIUS/nm | 0.031 | 0.065 | 0.099 | 0.113 | 0.135 | 0.140 |
| $1^{ST}$ I.E/$KJMol^{-1}$ | 900 | 736 | 590 | 548 | 502 | |
| $2^{ND}$ I.E/$KJMol^{-1}$ | 1760 | 1450 | 1160 | 1060 | 966 | |
| MELTING POINT/$^oC$ | 1280 | 650 | 838 | 768 | 714 | |

The elements are alkaline earth metals which typically react by forming divalent $M^{2+}$ ions showing a well graded trend in reactivity which increases as atomic radius increases. Beryllium (Be) stands apart from the rest of the group mainly due to the small size of Be atom and the corresponding $Be^{2+}$ cation.

Fajan's rule states that ions with high charge densities have high polarising power and tend to form covalent compounds. Beryllium atom also has a high electronegativity such that the difference in the electronegativity when Be reacts with non-metals is seldom large (very small), as a result compounds such as $BeF_2$ and BeO show evidence of covalent character.

The increase in reactivity from Be to Ra is due to decrease in the effective nuclear charge of the atoms going down. Although the nuclear charge increases from one element to the next, the increase in screening effect of the inner electrons outweighs the increase in the nuclear charge such that effective nuclear charge decreases from one element to another. It becomes easier for the element to lose their valence electrons down the group.

Melting and boiling point decreases as the group is descended due to weakening of the metallic bonds as the atomic radius increases. The strength of metallic bonds is expected to decrease constantly down the group as effective nuclear charge decreases with increasing atomic radius. The trend in melting point is however irregular because the elements adopt different crystal structures (packaging atoms).

## CHEMICAL PROPERTIES OF THE ELEMENTS

❖ ***Reactions with Oxygen***

The elements are good reducing agents they all reacts with oxygen forming oxides as follows:

$$2M_{(s)} + O_{2(g)} \longrightarrow 2MO_{(s)}$$ :where M is group II metal

- **Beryllium** tarnishes in air forming a layer of white oxide on top of a grey metal. The reaction occurs rapidly that it is usually stored under oil like sodium. On heating in oxygen it forms a tough layer of BeO which protects the metal below from further reaction.

- **Magnesium** burns in air with an intense, very bright, white dazzling flame forming MgO. The combustion also produces $Mg_3N_2$ due to the large amount of energy produce which facilitates the cleavage of the N☰N for nitrogen to react.

- **Calcium** when exposed to air readily reacts with oxygen forming CaO on its surface. As a result of this increased reactivity Ca has to be stored under paraffin On igniting in air it gives a brick red flame.

- **Barium** reacts with oxygen with increasing reactivity. It burns in air with a pale green flame to form BaO and in excess oxygen it produces white $BaO_2$ (barium peroxide) in addition to BaO.

- **Strontium** combusts readily in oxygen giving a crimson flame.

❖ ***Reaction with Water***

Reactions becomes increasingly vigorous down the group with Be showing no reaction with water due to the formation of the tough protective oxide layer of BeO.

- **Magnesium:** cleaned magnesium ribbon reacts very slowly with cold water producing a few tiny bubbles of $H_{2(g)}$ and a slightly alkaline solution of $Mg(OH)_2$ of pH≈9 but rapidly with steam giving $MgO_{(s)}$ and $H_{2(g)}$.

As for the rest, they react with increasing rapidity with cold $H_2O_{(l)}$ producing a metal hydroxide and hydrogen gas as follows:

$$M_{(s)} + 2H_2O_{(l)} \longrightarrow M(OH)_{2(aq)} + H_{2(g)}$$ :where M is Ca, Ba or Sr

- **Calcium:** reacts steadily with cold water producing bubbles f $H_2$ gas and partially soluble precipitate of $Ca(OH)_2$. The resulting solution is alkaline with a pH≈11.
- **Barium:** reacts vigorously with cold water producing a large amount of $H_2$ gas and an alkaline solution of $Ba(OH)_2$ of pH≈13

❖ ***Reactions of the oxides with water***

All the oxides are partially soluble in water and yield an alkaline solution according to the general equation below:

$$MO_{(s)} + H_2O_{(l)} \longrightarrow M(OH)_{2(aq)}$$ :where M is group II metal

$$\Delta H_{soln} = -LE + \Delta H_{hyd}$$

Down the group lattice energy (LE) of the oxides decreases due to increasing ionic radius of the cations therefore solubility of the oxides increases as less energy is required to break the lattice thus the forces of attraction between the

cation and the $O^{2-}$ anion decreases, hence BeO and MgO are insoluble, CaO, BaO and SrO shows an increasing solubility.
The resultant hydroxides become increasingly soluble down the group with $Mg(OH)_2$ being insoluble, $Ca(OH)_2$ sparingly soluble and $Ba(OH)_2$ soluble.
The basicity of the oxides increases down the group showing a decrease in covalent character as polarising power of $M^{2+}$ cation decrease, therefore they react with acids to give a salt and water for the exception of BeO which is amphoteric.

- **CARBONATES AND NITRATES**

Group II elements form carbonates with general formula $MCO_3$ and nitrates with general formula $M(NO_3)_2$. The carbonates are all insoluble (precipitated as white precipitates) and the nitrates are all soluble in water.

**Thermal stability of Carbonates and Nitrates of group II elements**

Thermal stability refers to decomposition of the compounds on heating. Increased thermal stability means a higher temperature is needed to decompose the compound. The nitrates/carbonates of group II elements decompose to leave a metal oxide residue as follows:

**(a) For Carbonates:**

$MCO_{3(s)} \xrightarrow{\Delta H} MO_{(s)} + CO_{2(g)}$ : Where M is group II element

**(b) For Nitrates:**

$2M(NO_3)_{2(s)} \xrightarrow{\Delta H} 2MO_{(s)} + 4NO_{2(g)} + O_{2(g)}$

: Where M is group II element

| Carbonate | $MgCO_3$ | $CaCO_3$ | $SrCO_3$ | $BaCO_3$ |
|---|---|---|---|---|
| **Decomposition Temperature/ °C** | 550 | 900 | 1100 | 1400 |

- Thermal stability of the nitrates/carbonates increases down the group i.e more heat energy will be required to decompose the carbonates or nitrates on going down the group.
- The ease of decomposition is due to decrease in polarising power of the $M^{2+}$ cation on going down the group as charge density decreases with increasing ionic radius. The smaller ionic radii, the higher the charge density thus the higher the polarisation of the $NO_3^-/CO_3^{2-}$ by the $M^{2+}$ cation which weakens the N-O and the C-O bonds hence less energy will therefore be required to decompose the anions.

## ❖ SULPHATES AND HYDROXIDES OF GROUP II ELEMENTS

All group II elements form sulphates with general formula $MSO_4$.

### Solubility of Group II Sulphates

Solubility of ionic compounds is determined by lattice and hydration enthalpies.

$$MSO_{4(s)} \xrightarrow{\Delta H_{soln}} M^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)}$$

$$MSO_{4(s)} \xrightarrow{\Delta H_{latt}} M^{2+}_{(g)} + SO_4^{2-}{}_{(g)} \xrightarrow{\Delta H_{hyd}} M^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)}$$

$$\Delta H_{soln} = -\Delta H_{latt} + \Delta H_{hyd}$$

The $-\Delta H_{latt}$ (reverse lattice) is always endothermic whereas the $\Delta H_{hyd}$ is exothermic. Going down the group $\Delta H_{hyd}$ becomes less exothermic as charge density of cation decreases with increasing ionic radius. At the same time the $-\Delta H_{latt}$ becomes less endothermic as ionic lattice weakens with decreasing charge density. However the decrease in exothermicity of $\Delta H_{hyd}$ occurs at a faster rate than in endothermicity of $-\Delta H_{latt}$. The small change in $-\Delta H_{latt}$ from element to element is due to the influence of the large $SO_4^{2-}$ ion which is far much larger than the cation.

$$\text{Lattice Energy} \propto \frac{|q+|.|q-|}{ra + rc} \qquad :\text{where } r_a \gg r_c$$

$r_a + r_c$ remains large down the group

As a result $\Delta H_{soln}$ becomes more endothermic on descending the group and solubility decrease. The very low solubility of $BaSO_4$ is used as a test for presence of $SO_4^{2-}$ ions in solution by the addition of a solution with $Ba^{2+}_{(aq)}$ ions (e.g $BaCl_2$ or $Ba(NO_3)_2$) giving a white ppt. Although barium compounds are poisonous, a barium meal containing a suspension of $BaSO_4$ in water is given to patients with digestive tract problems before an X-ray examination as it is insoluble. Barium absorbs X-ray well and will show an outline of the digestive tract on an X-ray.

### Solubility of Group II Hydroxides

| HYDROXIDES | $Mg(OH)_2$ | $Ca(OH)_2$ | $Ba(OH)_2$ | $Sr(OH)_2$ |
|---|---|---|---|---|
| SOLUBILITY g/100cm³ | Sparingly soluble $2.0 \times 10^{-3}$ | Soluble $1.6 \times 10^{-3}$ | Very soluble $2.4 \times 10^{-2}$ | Very soluble $3.3 \times 10^{-2}$ |

$$M(OH)_{2(s)} \xrightarrow{\Delta H_{soln}} M^{2+}_{(aq)} + 2OH^-_{(aq)}$$

$$M(OH)_{2(s)} \xrightarrow{\Delta H_{latt}} M^{2+}_{(g)} + 2OH^-_{(g)} \xrightarrow{\Delta H_{hyd}} M^{2+}_{(aq)} + 2OH^-_{(aq)}$$

The solubilities of Group II hydroxides surprisingly show an opposite trend to that of the sulphates. The solubility of $M(OH)_2$ actually increases down the group. $\Delta H_{hyd}$ becomes less exothermic so as $-\Delta H_{latt}$, however the $-\Delta H_{latt}$ changes at a faster rate since the anion is small as a result $\Delta H_{soln}$ becomes more exothermic hence solubility of the hydroxides increase down the group. For this reason, $Mg(OH)_2$ is used as an anti-acid stomach tablets and in toothpaste as use of more soluble hydroxides such as $Ca(OH)_2$ would make stomach pH to be too alkaline as to have detrimental effect.

**USES OF SOME GROUP II ELEMENTS AND THEIR COMPOUNDS**

1. **Beryllium:** alloyed with copper to produce hard, electrical conductivity and corrosion resistant alloys.

BeO- due to its high melting point and stability it is used as a refractory material e.g spark plug insulator or space crafts

2. **Magnesium:** (Refer to period 3)
3. **Calcium:** extraction of uranium and as a de-oxidant in metal casings
   *$Ca(OH)_2$ :-* lime water solution, slacked lime (saturated suspension) used to increase pH of acidic soils,
   *$CaCO_3$* :- limestone for neutralising acidic soils, manufacturing cement, extraction of iron, toothpaste
   *$CaCl_2$ :-* used to lay dust on roads, drying agent for gases
   *$CaSO_4$ :-* making plaster for in buildings, plaster casts for broken limps, gypsum used to treat soil after flooding with sea $H_2O$ to render it fertile and workable
   *CaO :-* quick lime- for neutralising acidic soils, smelting of metal ores to remove silicate impurities, manufacture of glass and refractory material for furnace lining.

# GROUP IV ELEMENTS

Elements in Group IV are carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb). All have an outer configuration of $ns^2\ np^2$. The major feature in this group is the change from non-metallic C and Si through semi-metal germanium, to metallic Sn and Pb.

## VARIATION IN PHYSICAL PROPRTIES OF THE ELEMENTS

| ELEMENTS | C | Si | Ge | Sn | Pb |
|---|---|---|---|---|---|
| TYPE | Non-Metal | | Metalloid | Metals | |
| STRUCTURE AND BONDING | Giant Molecular | | | Giant Metallic | |
| MELTING POINT/°C | 3600[d.] 3750[g.] | 1410 | 937 | 232 | 327 |
| IONISATION ENERGY/ KJMol$^{-1}$ | 1090 | 786 | 762 | 707 | 716 |
| CONDUCTIVITY | Fairly good[g.] Nil[d.] | Semi-conductor | | Very good conductor | |

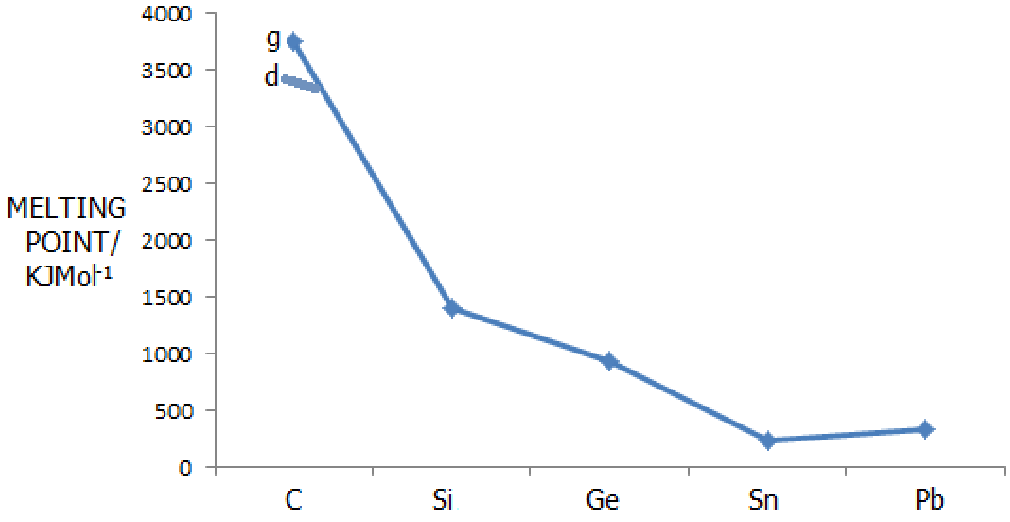
g- graphite d- diamond

❖ ***Metallic and Non-Metallic Structure***

- Going down the group elements change from being non-metals (carbon and silicon) through metalloid (germanium) to metallic (Sn and Pb).
- The first two elements carbon and silicon have relatively smaller atomic sizes and higher electronegativity. This results in the elements having higher ionisation energies and thus typically reacts through the formation of covalent bonds.
- As atomic radius increases, the electronegativity decreases resulting in lower ionisation energies hence they show more metallic behaviour.
- The bonding in the elements thus also changes from covalent to metallic.
- Carbon, Silicon and Germanium have giant molecular structures which are similar in shape and coordination to diamond. In these structures the atoms are linked by a network of covalent bonds running throughout the lattice.
- Tin and Lead have giant metallic structures in which metal atoms lose their valence electrons into the metallic lattice. Electrostatic forces between the positively charged metal atoms and delocalise electrons hold the metallic lattice together.

## ❖ ***Melting Point of Group IV elements***

Generally the melting points decrease going down the group.



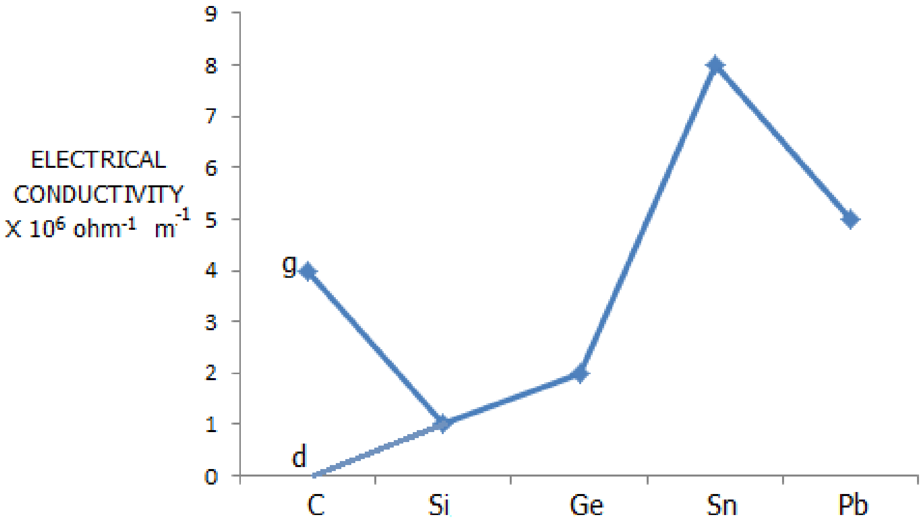

- C, Si and Ge have very high melting point compared to Sn and Pb due to the large amount of energy required to break the network of covalent bonds in the giant molecular stricter. However their melting points decrease gradually from carbon to germanium as covalent bonds become weaker with increasing bond length.
- For Carbon, diamond has slightly lower melting point than graphite because the carbon to carbon bond in graphite is shorter and stronger as they consists of sigma and pi bonds resulting from electron delocalisation whereas in diamond it is purely sigma.
- Tin and Lead are giant metallic structures which are held by fairly strong metallic bonding hence their lower melting point. The melting point of Pb is higher than that of Sn due to greater effective nuclear charge in Pb which results from poor shielding effect of 4f orbital. The Pb nucleus thus attracts the delocalised electrons more strongly forming stronger metallic bonding.

## ❖ ***First Ionisation Energy***

- As expected the first ionisation values of Group IV elements generally decrease on going down the group. As atomic radius increases, distance of outer electrons from nucleus increases at the same time shielding effect increases as number of electrons between outermost electron and nucleus increases.
- Although number of protons increases from one element to another the increase in the nuclear charge is outweighed by screening effect such that the effective nuclear charge decreases from one element to the other. Thus less energy is required to remove the outer shell electrons going down the group.
- The first ionisation energy decreases considerably from carbon to silicon. After that it falls relatively little because after silicon there is a large increase in nuclear charge which is associated with filling of the d and f orbitals, counterbalancing the increase in atomic radius.
- The anomaly between SN and Pb is again due to the poor shielding effect of 4f orbitals in Pb such that effective nuclear charge of Pb is greater than that of Sn.

## ❖ ***Electrical Conductivity***

- Carbon is a fairly good electrical conductor in the form of graphite due to the presence of delocalised electrons along the hexagonal layer and in the form of diamond it is a non-conductor due to the absence of delocalised electrons.
- Silicon and Germanium are semi-conductors and have lower electrical conductivities than graphite. In semi-conductors the electrons in the highest occupied molecular orbital (HOMO) can be promoted into the conduction band if given enough energy. This explains why electrical conductivity of semi-conductors increases with temperature whereas that of metallic conductors decreases with temperature.
- Tin and Lead are very good conductors due to the presence of delocalised valence electrons in the metallic lattice. Lead has a slightly lower conductivity due to some covalence nature as a result of the large atomic size.




## ❖ ***Group IV Tetra-chlorides $XCl_4$***

Overall reactivity increases down the group, all the elements react directly with chlorine gas to form tetra-chlorides of the formula $XCl_4$.

- $CCl_4$ is more easily prepared by reaction of $CS_2$ (carbon disulphide) and $Cl_2$.

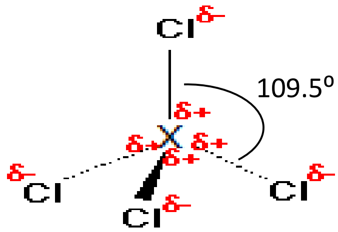
$$CS_2 + 3Cl_2 \longrightarrow CCl_4 + S_2Cl_2$$

- $PbCl_4$ prepared by action of conc. HCl on $PbO_2$ in the cold

$$PbO_2 + 4_{conc}HCl \longrightarrow PbCl_4 + 2H_2O$$

All the Group IV tetra-chlorides are simple molecular structures in which the $XCl_4$ molecules are held by Van der Waals' forces. They are all volatile liquids with low boiling point due to weak Van der Waals' forces.

The volatility decreases going down the group as number of electrons in $XCl_4$ molecules increase. Having 4 bonding pairs and zero lone pairs in the valence shell of the Group IV element, the tetra-chlorides are all tetrahedral in shape with a bond angle ≈109.5°.




The thermal stability of the tetra-chlorides decreases down the group as:

(a) The covalent bonds become weaker as the group IV atom become larger

(b) The inert pair effect increases down the group hence the +4 oxidation state becomes less stable: $CCl_4$, $SiCl_4$ and $GeCl_4$ are thermally stable to high temperature. $SnCl_4$ decomposes on heating to give $SnCl_2$ and $Cl_2$. $PbCl_4$ a yellow liquid decomposes to $PbCl_2$ (white solid) and $Cl_{2(g)}$ at r.t.p

**Hydrolysis of tetrachlorides**

All the Group IV tetra-chlorides except $CCl_4$ are easily hydrolysed by water forming the respective hydroxyl compounds or hydrated oxides and HCl.

$$XCl_{4(l)} + 4H_2O_{(l)} \longrightarrow X(OH)_{4(aq)} + 4HCl_{(aq)}$$

e.g $SiCl_{4(l)} + 4H_2O_{(l)} \longrightarrow Si(OH)_{4(aq)} + 4HCl_{(aq)}$

$$SiCl_{4(l)} + 4H_2O_{(l)} \longrightarrow SiO_2.2H_2O_{(s)} + 4HCl_{(aq)}$$

$CCl_4$ is resistant to hydrolysis because, during the hydrolysis reaction the oxygen atom in water donates its lone pair of electrons into the vacant d-orbital there-by weakening the X – Cl bonds, but Carbon being a period 2 element, its d-orbitals are far high in energy and are inaccessible in terms of energy. Furthermore the C atom being small, with the 4 Cl atoms clustered around it offers steric hindrance to the hydrolysing water molecules, inhibiting the hydrolysis reaction.

The rate of hydrolysis reaction increases from $SiCl_4$ to $PbCl_4$, as the X-Cl bond becomes weaker with increase in both lengths.

- ***Group IV Oxides***

  All group IV elements form oxides in which they are in +2 or +4 state. The stability and type of bonding present differs in these oxides.

**+2 Oxidation state**

| OXIDE | CO | SiO | GeO | SnO | PbO |
|---|---|---|---|---|---|
| **STRUCTURE** | Simple molecular | | Intermediate but increasingly ionic towards PbO | | |
| **THERMAL STABILITY** | Readily oxidised to dioxides | Form dioxides on standing in air | | | Stable |
| **ACID-BASE NATURE** | Neutral oxides react with neither acids nor alkalis | | Amphoteric oxides | | |

**+4 Oxidation State**

| OXIDE | $CO_2$ | $SiO_2$ | $GeO_2$ | $SnO_2$ | $PbO_2$ |
|---|---|---|---|---|---|
| STRUCTURE | Simple molecular | Giant molecular | Intermediate between giant ionic and covalent | | |
| THERMAL STABILITY | Stable even at high temperatures | | | | Decomposes to PbO on warming |
| ACID-BASE NATURE | Acidic | | Amphoteric oxides | | |

**Carbon:-** forms CO (+2) and $CO_2$ (+4) both of which are simple molecular. CO is neutral but $CO_2$ is acidic reacting with alkalis forming carbonate salt and water.

$$CO_2 + 2OH^- \longrightarrow CO_3^{2-} + H_2O$$

Of the two, $CO_2$ is far the more stable, CO on standing in air is easily oxidised to $CO_2$

$$2CO + O_2 \longrightarrow CO_2$$

$CO_2$ is thermally stable and does not decompose even on heating to very high temperatures.

**Silicon:-** forms SiO (+2) which is simple molecular and $SiO_2$ (+4) which is giant molecular. SiO is very unstable and only exist at very high temperatures. SiO is neutral whereas $SiO_2$ is acidic reacting with alkalis forming silicate salts and water.

$$SiO_2 + 2OH^- \longrightarrow SiO_3^{2-} + H_2O$$

**Ge, Sn and Pb:-** form +2 oxides which are predominantly ionic but have a significant covalent nature. Their +4 oxides are intermediate between giant molecular and ionic. For Ge and Sn the +4 oxidation state appears to be more stable as it does not decompose even on heating to very high temperatures whereas $PbO_2$, readily decompose on warming forming +2 oxide.

$$2PbO_2 \xrightarrow{\Delta} 2PbO + O_2$$

For Ge, Sn and Pb both +2 and +4 oxides are amphoteric because of the mixed type of bonding (ionic + covalent) and thus react with both acids and bases.

***The +2 oxides react with acids giving +2 salts and water as follows:***

$$SnO_{(s)} + 2H^+_{(aq)} \longrightarrow Sn^{2+}_{(aq)} + H_2O_{(l)}$$
$$PbO_{(s)} + 2H^+_{(aq)} \longrightarrow Pb^{2+}_{(aq)} + H_2O_{(l)}$$

***The +2 oxides react with alkalis forming +2 tri-hydroxo salts as follows:***

$$PbO_{(s)} + OH^-_{(aq)} + H_2O_{(l)} \longrightarrow Pb(OH)_3^-{}_{(aq)}$$ (tri-hydroxoplumbate (II))

$$SnO_{(s)} + OH^-_{(aq)} + H_2O_{(l)} \longrightarrow Sn(OH)_3^-{}_{(aq)}$$ (tri-hydroxostannate (II))

***The +4 oxides react with acids giving +4 salts and water as follows:***

$$PbO_{2(s)} + 4H^+_{(aq)} \longrightarrow Pb^{4+}_{(aq)} + 2H_2O_{(l)}$$
$$SnO_{2(s)} + 4H^+_{(aq)} \longrightarrow Sn^{4+}_{(aq)} + 2H_2O_{(l)}$$

***The +4 oxides react with fused or conc. alkalis forming $XO_3^{2-}$ salts***

$$GeO_{2(s)} + 2OH^-_{(aq)} \longrightarrow GeO_3^{2-}{}_{(aq)} + H_2O_{(l)} \quad \text{(germanate (IV))}$$
$$PbO_{2(s)} + 2OH^-_{(aq)} \longrightarrow PbO_3^{2-}{}_{(aq)} + H_2O_{(l)} \quad \text{(plumbate (IV))}$$
$$SnO_{2(s)} + 2OH^-_{(aq)} \longrightarrow SnO_3^{2-}{}_{(aq)} + H_2O_{(l)} \quad \text{(stannate (IV))}$$

Therefore in general the following trends are seen in the oxides of group IV elements:

1. The +4 oxide becomes less stable down the group as +2 oxide increases in stability. By the time we reach Pb the +2 becomes the most stable state.
2. The bonding in the oxides particularly +4 oxides changes from simple molecular through giant molecular to ionic, that is ionic character in oxide increases down the group
3. The oxides, particularly +4 oxides changes from acidic to amphoteric.

## ❖ ***Relative Stability of the +2 and +4 oxidation states***

As observed with the oxides the +4 oxidation state becomes less stable for group IV compounds on descending the group and the +2 becomes more stable.

- In **C** and **Si** compounds, the +4 is very stable relative to the +2 and +2 compounds are rare and easily oxidised to +4 state.
  As observed that CO reacts very readily and exothermically to form $CO_2$. SiO is very instable and it is obtained only at temperatures as high as 2000°C.
- For **Ge** +4 is still more stable relative to +2 oxidation state. By the time we reach Pb, the +2 is now more stable relative to +4. $PbO_2$ is strongly oxidising and can oxidise concentrated HCl to $Cl_2$.

$$PbO_{2(s)} + 4_{conc}HCl \longrightarrow PbCl_{2(aq)} + Cl_{2(g)} + 2H_2O_{(l)}$$

- In Sn compounds the +4 oxidation state is only slightly more stable than 2 states, such that Sn (II) ions are only weak reducing.
- The relative stability of +2 and +4 oxidation state can also be shown by the trend in the standard electrode potential values.

$$Ge^{4+} + 2e^- \longrightarrow Ge^{2+} \qquad E^\theta = -1.06V$$
$$Sn^{4+} + 2e^- \longrightarrow Sn^{2+} \qquad E^\theta = +0.15V$$
$$Pb^{4+} + 2e^- \longrightarrow Pb^{2+} \qquad E^\theta = +1.69V$$

The standard electrode potentials get more positive down the group showing that the stability of +4 oxidation state decreases relative to +2 and the oxidising power of +4 state increases going down the group.
The decrease in stability of +4 state accompanied by an increase in stability of +2 state can be explained by **inert pair effect.**

Group IV elements have outer configuration $ns^2$ $np^2$, the two's' electrons remain relatively stable and become less readily available for bonding. These electrons remain paired and do not participate in bonding. If the energy required to unpair them exceeds the energy evolved when they form bonds, then they remain paired and are not used in bonding.
This energy for unpairing the electrons tends to increase on going down the group hence the increase in their inertness. The elements will now form compounds using mainly the $np^2$ electrons. The inert pair effect is not restricted to Group IV elements only. It also explains the existence of +1 compound in the lower Group III elements such as Thallium (Tl), Indium (In) and Gallium (Ga) where higher elements in the group such as aluminium are restricted to +3 states.

## Uses of Group IV elements and their Compounds

### Carbon

***Graphite***

1. Used as electrodes in most electrolytic cells because it is an inert electrode with higher melting point during electrolysis of $Al_2O_3$.
2. Used to make exhaust cones for rockets and ladles due to its higher melting point.
3. Used as a lubricant because it is slippery.

***Diamond***

1. Used as an abrasive for smoothening hard surfaces and drill tips used in mining because of its hard lattice structure.
2. Jewellery because has a high refractive index

$CO_2$

1. Dissolved in carbonated drinks under high pressure, acts as a preservative (excludes air) and has a refreshing taste.
2. As a refrigerant as dry ice sublimes at -78°C
3. Used in fire extinguisher to put our fires because it displaces air since it is denser.
4. Making urea $NH_2CONH_2$ a nitrogenous fertiliser.

***Silicon***

1. Very pure Si a semiconductor is used in electronics to make transistors.
2. $SiO_2$ used in the manufacture of glass and cement.
3. $SiO_2$ combined with carbon to form carborandum (CSi) used as an abrasive.

***Germanium***

Very pure Ge used as a semi-conductor in electronics.

***Tin***

1. Principally used in coating steel to prevent corrosion.
2. Used in food and drink canning because Sn is not poisonous.
3. Manufacture of soldering alloy
4. $SnO_2$ used to make white enamels and tiles.

***Lead***

1. Element used in plumbing, roofing and for sheathing cables due to its resistance to corrosion and low melting point.
2. $PbO_2$ and Pb used as electrodes in car batteries
3. Used in soldering alloys usually mixed with Sn.
4. Pb bricks are used to build screening walls around nuclear reactors, X-ray generators and other powerful sources of radiation. Pb also used for transporting radioactive isotopes.
5. PbO used in manufacture of lead glass.

# GROUP VII- HALOGENS

The name **halogen** comes from Greek, and it means 'salt formers'. All halogens react with metals to form salts. The elements all have 7 electrons in their valence shell with configuration $ns^2$ $np^5$.

They complete their octet by either accepting one electron forming halide ions or by sharing an electron with other elements in covalent bonding. Their compounds with metals are typically ionic and with non-metals are covalent. The halogens show very close group similarities but F the first member shows deviation from the rest in number of ways.

The first element of each of the main groups all show differences from subsequent elements due to the following reason:

1. The first element is smaller than others and holds its electrons more firmly that is they have high electronegativity.
2. It has no low lying d-orbitals which may be used for bonding.

## ❖ PHYSICAL PROPERTIES

| ELEMENT | Fluorine | Chlorine | Bromine | Iodine |
|---|---|---|---|---|
| OUTER SHELL ELECTRON CONFIGURATION | $2s^2$ $2p^5$ | $3s^2$ $3p^5$ | $4s^2$ $4p^5$ | $5s^2$ $5p^5$ |
| STATE @ R.T.P | Gas | | Liquid | Solid |
| COLOUR | Pale Yellow | Pale Green | Red-brown | Dark grey/Shiny black solid (Purple in vapour state) |
| ATOMIC RADIUS/nm | 0.072 | 0.099 | 0.114 | 0.133 |
| IONIC RADIUS/nm | 0.136 | 0.181 | 0.195 | 0.216 |
| MELTING POINT/°C | -220 | -101 | -7 | 113 |
| BOILING POINT/°C | -188 | -35 | 59 | 183 |
| ELECTRON AFFINITY | 4.00 | 2.85 | 2.75 | 2.20 |
| SOLUBILITY IN WATER | Reacts readily with water | Moderately soluble | Slightly soluble | Insoluble |

In the elemental form the halogens are all diatomic ($hal_2$) simple molecular structures in which the discrete halogen molecules are held by weak Van der Waals forces. Consequently they have generally low melting points and boiling points due to weak Van der Waals forces. The strength of these forces increases going down the group due to increase in the number of electrons in the molecule resulting in the formation of stronger dipoles. As a result the volatility of the halogens decrease going down the group.

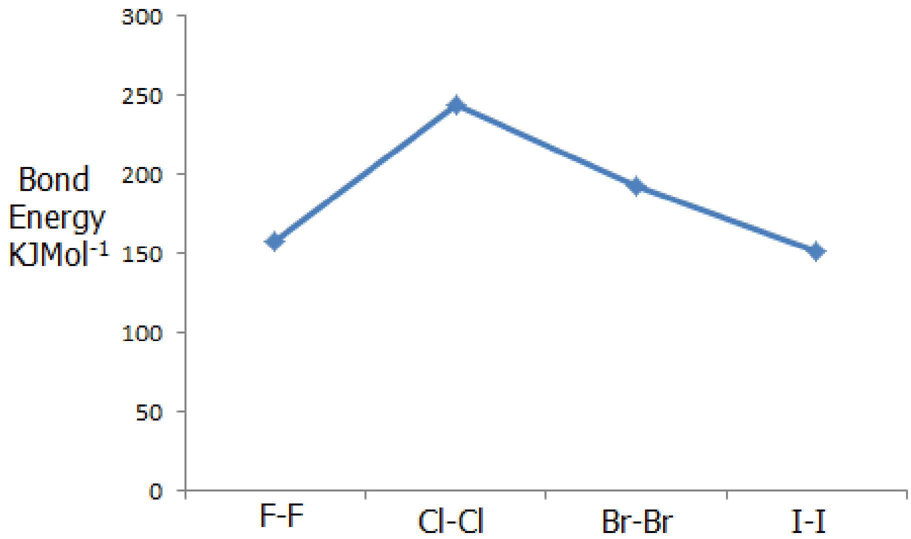
The colour intensity of the halogens also increases as we go down the group. The colour arises from the absorption of ultra-violet range and subsequent promotion of an electron in ground state to higher energy state. On descending the group the energy levels become

closer, so the promotion energy becomes less and wavelength becomes longer.
The halogens are soluble in organic solvents such as tetra-chloromethane, giving yellow $Cl_2$, red $Br_2$ and purple $I_2$ solutions.

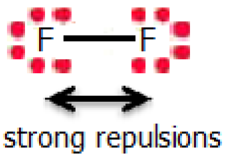
## ❖ ***Bond Energy of Halogens***

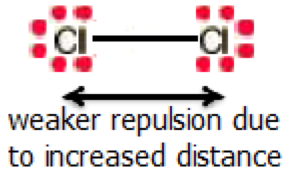
**Def:-** is the minimum amount of energy required to break one mole of a specific covalent bond

| Halide - halide | F-F | Cl-Cl | Br-Br | I-I |
|---|---|---|---|---|
| **Bond Energy / $KJMol^{-1}$** | 158 | 244 | 193 | 151 |




- The bond energy values increase from F-F to Cl-Cl and then gradually decrease from Cl-Cl to I-I. The decrease from $Cl_2$ to $I_2$ can be expected from the increasing bond length as atomic radius increase thus the shorter the bond length stronger the bond.
- The unusually low bond energy for such a short bond length in the F-F molecule arises from strong repulsions between the 3 lone pairs on each of the small F atoms, which are very close to each other weakening the short F-F bond.




This very low bond dissociation energy coupled with its high electronegativity causes $F_2$ to be the most reactive known element.

## ❖ ***Formation of Hydrogen Halides***

All halogens react with $H_2$ to form hydrogen halides but the rate of reaction decreases as reactivity of the halogens decreases down the group.

$$H_{2(g)} + X_{2(g)} \longrightarrow 2HX_{(g)}$$ where X: F, Cl, Br and I

- $F_2$ explodes in $H_2$ in the dark at a room temperature at which $F_2$ is solid.

$$F_{2(g)} + H_{2(g)} \longrightarrow 2HF_{(g)}$$

- $Cl_2$ will not react with $H_2$ at room temperature in the dark but reacts explosively in sunlight or ultra-violet light

$$Cl_{2(g)} + H_{2(g)} \longrightarrow 2HCl_{(g)}$$

- $Br_2$ will react with $H_2$ on heating in the presence of a platinum catalyst at 200°C

$$Br_{2(g)} + H_{2(g)} \longrightarrow 2HBr_{(g)}$$

- $I_2$ react with $H_2$ at ≈ 400°C in the presence of Pt catalyst, but the reaction is reversible resulting in an equilibrium mixture.

$$I_{2(g)} + H_{2(g)} \Longleftrightarrow 2HI_{(g)}$$

The decrease in the reactivity is due to decrease in the H-X bond energy as bond length increases due to increase in the atomic radius of the halogen.

| H-X | H-F | H-Cl | H-Br | H-I |
|---|---|---|---|---|
| Bond Energy / $KJMol^{-1}$ | 562 | 431 | 366 | 299 |

$$H_{2(g)} + X_{2(g)} \longrightarrow 2HX_{(g)}$$

$\Delta H_{reaction} = \sum$bonds broken - $\sum$bonds formed

$= [BE_{(H-H)} + BE_{(X-X)}] - 2[BE_{(H-X)}]$

$\Delta H_{reaction}$ (HF) = -530 $KJMol^{-1}$

$\Delta H_{reaction}$ (HCl)= -182 $KJMol^{-1}$

$\Delta H_{reaction}$ (HBr)= -103 $KJMol^{-1}$

$\Delta H_{reaction}$ (HI) = -11 $KJMol^{-1}$

Enthalpy change of reaction become less exothermic down the group and reaction becomes less energetically feasible.

## Properties of Hydrogen Halides

### *Thermal Stability of H-X*

$$2HX_{(g)} \longrightarrow H_{2(g)} + X_{2(g)}$$

As expected from the decreasing bond enthalpies of H-X, their thermal stabilities also decrease down the group. On heating the H-X decomposes giving $H_2$ and $X_2$.
HCl and HF remain un-decomposed even up to very high temperatures. HI with lowest bond energy dissociates on heating being about 30% dissociation at 2000°C. HBr is only 1% dissociation at this temperature.

### *Acidity of Hydrogen Halogens*

The H-X is essentially covalent in the gas state, however in aqueous solution they ionise giving $H^+$ ions.

$$HX_{(g)} + H_2O_{(l)} \longrightarrow H_3O^+_{(aq)} + X^-_{(aq)}$$

The three acids hydrochloric, hydrobromic and hydroiodic acid are all acids and ionise almost completely. The order of increase in acidity is HF<HCl<HBr<HI. This can be expected from decreasing bond enthalpy of H-X bond which is more easily broken down the group.

The H-F bond is very strong (BE 562KJMol$^{-1}$) compared to other H-X bonds due to large difference in electronegativity and also the shorter bond. As a result HF is very weak acid only ≈ 9% ionised in 0.1M solution.

- ***Boiling Points of H-X***

| H-X | H-F | H-Cl | H-Br | H-I |
|---|---|---|---|---|
| **Boiling Point /°C** | +19.9 | -85.0 | -66.7 | -35.4 |

The HX are simple molecular compounds with low boiling points. The boiling points of HX decrease sharply from HF to HCl and gradually increase from HCl to HI. The gradual increase from HCl to HI is due to increase in the strength of Van der Waals forces between the HX molecules as number of electrons increases in the HX molecule.
HF has an unusual high boiling point for such a small molecule due to existence of hydrogen bonds between the HF molecules. The hydrogen bonds arise from the high electronegativity of the F which causes it to draw electrons strongly from H atom resulting in H acquiring a large partial positive charge and itself a large partial negative charge. Strong dipole-dipole attraction –hydrogen bonds are then formed between the $H^{\delta+}$ and $F^{\delta-}$.

## ***Halogens as Oxidising Agents***

Halogens react typically by accepting an electron forming the halide ion.

$$X_2 \quad + \quad 2e^- \longrightarrow 2X^-$$

- In doing so they will be acting as oxidising agents (*an oxidising agent is a substance which accepts electrons from another substance).* The halogens are very strong oxidising agents with the oxidising power decreasing down the group due to increase in atomic size and decrease in electron affinity.
- This is shown by their $E^\theta$ values of the halogens:

$$F_2 \quad + \quad 2e^- \longrightarrow 2F^- \quad E^\theta = +2.87$$

$$Cl_2 \quad + \quad 2e^- \longrightarrow 2Cl^- \quad E^\theta = +1.36$$

$$Br_2 \quad + \quad 2e^- \longrightarrow 2Br^- \quad E^\theta = +1.07$$

$$I_2 \quad + \quad 2e^- \longrightarrow 2I^- \quad E^\theta = +0.54$$

- The standard electrode potential values become less positive down the group showing decrease in oxidation power of the halogens. $F_2$ is the strongest oxidising agent of the halogens and indeed the strongest oxidising agent known. $F_2$ will oxidise all the other halogens as follows:

$$F_2 \quad + \quad 2X^-_{(aq)} \longrightarrow 2F^-_{(aq)} + \quad X_2$$

- In general any halogen high up in the group will oxidise halide ions of elements below it in the group (displacement reactions).

- The trend in the oxidising power can be explained in terms of the energy changes involved in the reaction:

$$\tfrac{1}{2}X_{2(standard)} + e^- \longrightarrow 2X^-_{(aq)}$$

The halogen molecules are first atomised to give gaseous atoms which then accept electrons to form gaseous ions. Since most of these occur in aqueous solution the gaseous halide ions are then hydrated by solvating $H_2O$ molecules.

$$\begin{tikzcd}
\tfrac{1}{2}X_{2(standard)} \arrow[r, "\Delta H_{reaction}"] \arrow[d, "\Delta H_{atomisation}"] & X^-_{(aq)} \\
X_{(g)} \arrow[r, "\Delta H_{EA}"] & X^-_{(g)} \arrow[u, "\Delta H_{hydration}"']
\end{tikzcd}$$

$$\Delta H_{reaction} = \Delta H_{atomisation} + \Delta H_{EA} + \Delta H_{hydration}$$

The corresponding values of the halogens are:

| | F | Cl | Br | I |
|---|---|---|---|---|
| **$\Delta H_{at}$ / KJMol$^{-1}$** | 79 | 121 | 112 | 107 |
| **$\Delta H_{EA}$ / KJMol$^{-1}$** | -333 | -364 | -342 | -295 |
| **$\Delta H_{hyd}$ / KJMol$^{-1}$** | -457 | -381 | -351 | -307 |
| **$\Delta H_{rxn}$ / KJMol$^{-1}$** | -711 | -624 | -581 | -495 |

- The enthalpy changes for the reaction in which halogens act as oxidising agents become less exothermic down the group, thus it becomes less energetically favourable for the halogens to act as oxidising agents down the group.
- The decrease in exothermicity of reaction is mainly due to:
  - Decrease in $\Delta H_{hyd}$ as ionic radius increases hence charge density of $X^-$ increases.
  - Electron affinity for the halogens also becomes less exothermic as atomic radius increases.
- The high oxidising power of $F_2$ is mainly due to:
  - Its low enthalpy of dissociation arising from weakness of the F-F bond.
  - Its high energy of hydration arising from the small size of the $F^-$ ion.

***The oxidising ability of the halogens can be illustrated by the following:***

**<u>Reactions with thiosulphate ions $S_2O_3^{2-}$</u>**

$F_2$, $Cl_2$ and $Br_2$ react with thiosulphate ions oxidising them to sulphate ions

$$4X_{2(aq)} + \underset{(+2)}{S_2O_3^{2-}{}_{(aq)}} + 5H_2O_{(l)} \longrightarrow 8X^-_{(aq)} + \underset{(+6)}{2SO_4^{2-}{}_{(aq)}} + 10H^+_{(aq)}$$

The oxidation state of S increases from +2 to +6 that is by +4 units.

$I_2$ the most weakly oxidising of the halogens, oxidises $S_2O_3^{2-}$ to tetrathionate ions $S_4O_6^{2-}$.

$$I_2 + \underset{(+2)}{S_2O_3^{2-}{}_{(aq)}} \longrightarrow \underset{(+2.5)}{S_4O_6^{2-}{}_{(aq)}} + 2I^-_{(aq)}$$

The oxidation state increases from +2 to +2.5, showing that $I_2$ is a weak oxidising power.

### ❖ Reaction of halide with concentrated sulphuric acid

Concentrated sulphuric acid is a mild oxidising agent. All halide ions react with concentrated sulphuric acid initially producing white fumes of hydrogen halide. With chloride ions the reaction does not go further.

$$Cl^-_{(aq)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(White fumes)}}{HCl_{(g)}} + HSO_4^-{}_{(aq)}$$

With $I^-$ and $Br^-$ the HX produced is oxidised by concentrated $HSO_4$ to $I_2$ and $Br_2$ respectively.

$$Br^-_{(aq)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(White fumes)}}{HBr_{(g)}} + HSO_4^-{}_{(aq)}$$

$$2HBr_{(g)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(Red brown fumes of }Br_2)}{Br_{2(g)}} + 2H_2O_{(l)} + SO_{2(g)}$$

$$I^-_{(aq)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(White fumes)}}{HI_{(g)}} + HSO_4^-{}_{(aq)}$$

$$2HI_{(g)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(Black solid \& purple fumes)}}{I_{2(g)}} + 2H_2O_{(l)} + SO_{2(g)}$$

With $I_2$ the oxidation occurs to a greater extend with some $H_2SO_4$ reduced to $H_2S$.

$$I^-_{(aq)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(White fumes)}}{HI_{(g)}} + HSO_4^-{}_{(aq)}$$

$$8HI_{(g)} + H_2SO_{4(conc)} \longrightarrow \underset{\text{(Black solid \& purple fumes)}}{4I_{2(g)}} + 4H_2O_{(l)} + \underset{\text{(Pungent smell)}}{H_2S_{(g)}}$$

The above trend shows that concentrated $H_2SO_4$ can oxidise bromide and iodide ions to $Br_2$ and $I_2$ respectively but it is not sufficiently strongly oxidising to oxidise chloride to $Cl_2$ because $Cl_2$ is itself strongly oxidising.

### ❖ Reaction with $Fe^{2+}{}_{(aq)}$

$F_2$, $Cl_2$ and $Br_2$ can oxidise $Fe^{2+}$ ions to $Fe^{3+}$ ions as supported by positive $E^\theta$ values for the reactions.

$$F_{2(aq)} + 2Fe^{2+}{}_{(aq)} \longrightarrow 2F^-_{(aq)} + 2Fe^{3+}{}_{(aq)} \qquad E^\theta = +2.10$$

$$Cl_{2(aq)} + 2Fe^{2+}{}_{(aq)} \longrightarrow 2Cl^-_{(aq)} + 2Fe^{3+}{}_{(aq)} \qquad E^\theta = +0.59$$

$$Br_{2(aq)} + 2Fe^{2+}{}_{(aq)} \longrightarrow 2Br^-_{(aq)} + 2Fe^{3+}{}_{(aq)} \qquad E^\theta = +0.30$$

With iodine the iodine the reaction is not energetically feasible as shown by negative $E^\theta$ value.

$$I_{2(aq)} + 2Fe^{2+}{}_{(aq)} \longrightarrow 2I^-_{(aq)} + 2Fe^{3+}{}_{(aq)} \qquad E^\theta = -0.23$$

Instead iodine can be oxidised by $Fe^{3+}$ to $I_2$.

$$2I^-_{(aq)} + 2Fe^{3+}{}_{(aq)} \longrightarrow I_{2(aq)} + 2Fe^{2+}{}_{(aq)} \qquad E^\theta = +0.23$$

## Disproportionation reaction of halogens

A ***disproportionation reaction*** is a redox reaction in which the same species is reduced and oxidised in that reaction.

❖ ***Reaction with NaOH***

$Cl_2$, $Br_2$ and $I_2$ undergo disproportionation in alkalis. The products produced vary depending on the temperature of the solution.

In ***cold*** alkali the halogens produce a mixture of the halide and halate (I) ions.

$$X_2 + 2OH^-_{(aq)} \longrightarrow X^-_{(aq)} + XO^-_{(aq)} + H_2O_{(l)}$$

The $XO^-$ which is produced in the first reaction undergoes disproportionation at higher temperature.

$$\underset{+1}{3XO^-_{(aq)}} \longrightarrow \underset{-1}{2X^-_{(aq)}} + \underset{+5}{XO_3^-{}_{(aq)}} \text{ (halate (V) ion)}$$

The temperature at which disproportionation of $XO^-$ ion occurs decrease as we go down the group due to the decrease in the thermal stability of the $XO^-$ ion, such that the order of thermal stability is $ClO^- > BrO^- > IO^-$.

The reaction of $Cl_2$ with NaOH is employed in the production of bleaches. $NaClO_3$ is commonly used as a fabric bleaching agent. $Cl_2$ is bubbled in hot NaOH to produce NaCl and $NaClO_3$.

$$3Cl_{2(g)} + 6NaOH_{(aq)} \longrightarrow 5NaCl_{(aq)} + NaClO_{3(aq)} + 3H_2O_{(l)}$$

## Reaction with water

$$O_2 + 4H^+_{(aq)} + 4e^- \longrightarrow 2H_2O_{(l)} \qquad E^\theta = +1.23$$

$$F_2 + 2e^- \longrightarrow 2F^- \qquad E^\theta = +2.87$$

$$Cl_2 + 2e^- \longrightarrow 2Cl^- \qquad E^\theta = +1.36$$

The $E^\theta$ values show that $F_2$ and $Cl_2$ might be able to oxidise water to oxygen. $F_2$ is so strongly oxidising such that it oxidise water to $O_2$ and also ozone, hydrogen peroxide and $OF_2$.

$$2F_{2(g)} + 2H_2O_{(l)} \longrightarrow 4HF_{(g)} + O_{2(g)}$$

A similar reaction occurs between $Cl_2$ and water but the reaction is very slow due to high activation energy.

$$2Cl_{2(g)} + 2H_2O_{(l)} \longrightarrow 4HCl_{(aq)} + O_{2(g)}$$

Instead most of the dissolved $Cl_2$ undergoes a disproportionation reaction giving HCl and HClO.

$$Cl_{2(g)} + H_2O_{(l)} \longrightarrow HCl_{(aq)} + HClO_{(aq)}$$

On standing in sunlight the HClO gradually decomposes giving HCl and $O_2$.

$$2HClO_{(aq)} \longrightarrow 2HCl_{(aq)} + O_{2(g)}$$

The mechanism of reaction occurs through the production of oxygen free radicals. It is these oxygen radicals which kill bacteria and hence use $Cl_2$ in water treatment. The bleaching action of most $Cl_2$ is also due to formation of HClO, which oxidises organic dyestuffs to colourless substances. $Br_2$ and $I_2$ also dissolve and react with water in similar way with $I_2$ being least soluble.

## Reactions of Halide ions with $AgNO_3$

Silver nitrate, in the presents of dilute $HNO_3$ is commonly used to detect presence of $Cl^-$, $Br^-$ and $I^-$ ions by precipitation of silver halides. It will however not detect $F^-$ ions because AgF is surprisingly very soluble.
The AgX precipitates formed have varying solubility in $NH_{3(aq)}$ and this can be used to confirm presence of each halide ion.

- **Chloride** ions give a white precipitate with $AgNO_3$, which dissolves in dilute $NH_{3(aq)}$ due to formation of silver diamine complex ion $[Ag(NH_3)_2]^+$.

$$AgNO_{3(aq)} + Cl^-_{(aq)} \longrightarrow \underset{\text{(white ppt)}}{AgCl_{(s)}} + NO_3^-{}_{(aq)}$$

$$AgCl_{(s)} + 2NH_{3(aq)} \longrightarrow [Ag(NH_3)_2]^+_{(aq)} + Cl^-_{(aq)}$$

Complexation of $Ag^+$ by $NH_3$ ligands removes $Ag^+$ from the solubility equilibrium below shifting equilibrium to the right and hence AgCl precipitate dissolves.

$$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$$

$$Ag^+_{(aq)} + 2NH_{3(aq)} \longrightarrow [Ag(NH_3)_2]^+_{(aq)}$$

- **Bromide** ions give a cream precipitate of AgBr with $AgNO_3$. This precipitate dissolves only in concentrated $NH_3$.

$$Br^-_{(aq)} + AgNO_{3(aq)} \longrightarrow \underset{\text{(cream ppt)}}{AgBr_{(s)}} + NO_3^-{}_{(aq)}$$

$$AgBr_{(s)} + 2NH_{3(conc)} \rightleftharpoons [Ag(NH_3)_2]^+_{(aq)} + Br^-_{(aq)}$$

- **Iodide** ions give a yellow precipitate of AgI with $AgNO_3$ which is insoluble in both dilute and concentrated $NH_3$.

$$AgNO_{3(aq)} + I^-_{(aq)} \longrightarrow \underset{\text{(Yellow ppt)}}{AgI_{(s)}} + NO_3^-{}_{(aq)}$$

The silver halides show a similar trend in solubility with $Na_2S_2O_3$ or KCN owing again to the formation of soluble silver (I) complexes.

$$Ag^+_{(aq)} + 2S_2O_3^{2-}{}_{(aq)} \longrightarrow [Ag(S_2O_3)_2]^{3-}_{(aq)}$$

$$Ag^+_{(aq)} + 2CN^-_{(aq)} \longrightarrow [Ag(CN)_2]^-_{(aq)}$$

These two reactions are used in photography to remove from the film any AgX that has not been converted into Ag atoms by photodecomposition. The decrease in the solubility of AgX in $NH_{3(aq)}$ can be explained partly by looking at their $K_{SP}$ values.

| AgX | AgCl | AgBr | AgI |
|---|---|---|---|
| **$K_{SP}$ $mol^2dm^{-6}$ @ 25°C** | $2.0 \times 10^{-10}$ | $5.0 \times 10^{-13}$ | $8.0 \times 10^{-17}$ |

Precipitation occurs when ionic product Q is greater than $K_{SP}$. Going down the group the $K_{SP}$ values decrease, meaning that the solubility and hence $[Ag^+]$ free in solution decreases. On addition of $NH_3$ $[Ag(NH_3)_2]^+$ is formed which essentially reduces $[Ag^+]$ from the solubility equilibrium.
For AgCl, the $[Ag^+]$ is relatively high due to larger solubility such that on adding $NH_3$ the concentration of $Ag^+$ is significantly reduced such that $Q<K_{SP}$ and hence

precipitate dissolves. For AgI, there is low $[Ag^+]$ and also a very small $K_{SP}$ value such that the ionic product $Q>K_{SP}$ and precipitate does not dissolve.

**Uses of Halogens and their Compounds**

1. **$F_2$-** used in the manufacture of cryolite for extraction of aluminium
   **-**used in manufacture of plastics Poly-tetrafluoroethylene (PTFE) known as Teflon (inert solid plastic resistant to chemical attack), as well as CFCs (non-toxic refrigerants and aerosols)
2. **$Cl_2$-**used for purifying drinking and swimming water as it kill bacteria.
   - used to make organic solvents such as $CCl_4$, PVC, bleaching solutions, refrigerator fluids and aerosol propellants.
   -used for manufacture of DDT and BHC which are used as insecticides.
3. **$Br_2$-**used to make 1.2 dibromoethane added to lead petrol to remove Pb.
   -used to make alkyl bromides used as nematocides, pesticides and flame retardants. KBr used as a sedative and anticonvulsant for epilepsy and, AgBr in photographic films.
4. **$I_2$-**used to make iodo-form $CHI_3$ used as an antiseptic, used in diet in order to produce thyroxin, manufacture of medicines and dyes.

# NITROGEN & SULPHUR

## NITROGEN

Nitrogen is the most abundant gas in air ≈78% and it is obtained from fractional distillation of liquefied air. It is a colourless, odourless gas with a boiling point ≈ -196°C and is only slightly soluble in $H_2O$. It neither burns nor support combustion. $N_2$ consist of diatomic molecules which are extraordinarily stable, not undergoing dissociation at temperature less than 3000°C. These properties show that $N_2$ is a very inert gas due to strong N≡N triple bond which requires a lot of energy (BE= 994KJmol$^{-1}$) to be broken so as to form gaseous N atoms before it reacts.

Atmospheric $N_2$ however undergoes a few reactions under elevated conditions of temperature. Lightning and an electric arc provide enough energy to allow $N_2$ and $O_2$ in the atmosphere to combine to give NO.

$$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)} \qquad \Delta H = +180KJmol^{-1}$$

The reaction is endothermic and reversible but some of the NO combine with $O_2$ to make

$$2NO_{(g)} + O_{2(g)} \longrightarrow 2NO_{2(g)}$$

### Chemical Reactions of Nitrogen

❖ ***Nitrogen Oxides***

Nitrogen oxides are also produced in the internal combustion of engines where nitrogen and oxygen from air react due to the high temperatures in engines. These oxides NO and $NO_2$ will be present in exhaust fumes of vehicles and are responsible for much pollution by vehicles.

$$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)} \qquad \Delta H = +180KJmol^{-1}$$

As the gases are cooled the NO is oxidised to $NO_2$ therefore NO exists at high temperature.

$$2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)} \qquad \Delta H = -113KJmol^{-1}$$

**$NO_{X's}$ damage the environment in the following ways:**

(a) *Reacts with water in the atmosphere forming acid rain*

$$2NO_{2(g)} + H_2O_{(l)} \longrightarrow HNO_{2(aq)} + HNO_{3(aq)}$$

Nitrous acid is then oxidised to nitric acid then the overall equation:

$$4NO_{2(g)} + 2H_2O_{(l)} + O_{2(g)} \longrightarrow 4HNO_{3(aq)}$$

(b) *Reacts with other pollutants to form ozone which causes irritation to eyes.*

(c) *$NO_{X's}$ catalyses the oxidation of $SO_2$ to produce $SO_3$ which dissolve in $H_2O$ leading to the formation of $H_2SO_4$ (acid rain)*

$$SO_{2(g)} + NO_{2(g)} \longrightarrow SO_{3(g)} + NO_{(g)}$$

$$2NO_{(g)} + O_{2(g)} \longrightarrow 2NO_{2(g)}$$

$$SO_{3(g)} + H_2O_{(l)} \longrightarrow H_2SO_{4(aq)}$$

(d) *$NO_{X'S}$ also participate in depletion of the ozone*

$$O_{3(g)} + NO_{(g)} \longrightarrow O_{2(g)} + NO_{2(g)}$$

$$NO_{2(g)} + O_{(g)} \longrightarrow NO_{(g)} + O_{2(g)}$$

(e) Nitrogen oxides also react with un-burnt hydrocarbons from car exhausts forming photochemical smog. This is a haze which usually precipitates just above the ground in areas with dense traffic.

**Prevention of $NO_{X'S}$ Pollution**

Exhaust fumes from vehicles also contain another pollutant gas CO which is produced due to incomplete combustion of fuel. Modern cars are fitted with a catalytic converter which removes nitrogen oxides and CO simultaneously.

The catalytic converter is a labyrinth coated with a platinum-rhodium alloy which catalyses reduction of $NO_X$ to $N_2$ and oxidation of CO to $CO_2$.

$$2NO_{(g)} + 2CO_{(g)} \xrightarrow{\text{Pt-Rd}} N_{2(g)} + 2CO_{2(g)} \quad \text{OR}$$

$$2NO_{2(g)} + 4CO_{(g)} \xrightarrow{\text{Pt-Rd}} N_{2(g)} + 4CO_{2(g)}$$

In power stations $NO_{X'S}$ levels are reduced by spraying water onto the burning fuel.

***NB*:-** Lead free petrol must be used as lead poisons the platinum-rhodium catalyst making it ineffective. Catalytic converters are expensive because:

(a) The platinum used is costly

(b) Engines that use unleaded petrol are less efficient so more fuel is used.

## Nitrogen Compounds

### ***Ammonia ($NH_3$)***

(a) **Industrial Manufacture**

$NH_3$ is manufactured industrially by direct combination of $N_2$ and $H_2$ in the Haber-Bosch process. The $N_2$ and $H_2$ used must be thoroughly purified to remove impurities such as CO, sulphur compounds, $CO_2$ and $H_2O$ which react with and poison the catalyst. The reaction is exothermic therefore the following conditions are employed which gives a reasonable yield of $NH_3$ of about 17%.

***Conditions used:*** high pressure of 200-250 atm

Moderate temperature of 450-500 °C

Powdered iron catalyst

$$N_{2(g)} + 3H_{2(g)} \underset{450\text{-}500^oC}{\overset{200\ atm, Fe\ catalyst}{\Longleftrightarrow}} 2NH_{3(g)} \quad \Delta H = -92KJmol^{-1}$$

When equilibrium is reached the gases are passed into a cooling tower where the mixture is cooled and $NH_3$ condenses into liquid removing it from the gaseous equilibrium mixture. Unreacted $N_2$ and $H_2$ are recycled back into the reaction chamber, where they are mixed with fresh $N_2$ and $H_2$.

Recent technological advances, in the development of high strength alloys have allowed the use of pressures as high as 1000 atm as vessels can withstand the high pressures. This has significantly increased the yield of $NH_3$.

(b) **Laboratory Preparation of $NH_3$**

$NH_3$ is prepared by reaction of a strong alkali with an ammonium salt and then warming. $NH_3$ gas evolved turns moist/damp red-litmus blue.

$$NH_4^+{}_{(s)/(aq)} + NaOH_{(aq)} \xrightarrow{\text{warm/heat}} Na^+{}_{(aq)} + \uparrow NH_{3(g)} + H_2O_{(l)}$$ OR

heating a nitrate containing salt with Al powder in excess aqueous NaOH.

**Properties of Ammonia**

It is a colourless gas, boiling point -33.5°C with a strong sharp choking smell and is extremely soluble in water. The solubility of $NH_3$ is attributed to:

(a) Formation of H-bonds with solvating $H_2O$ molecules.

(b) Dissociation of $NH_3$ in aqueous solution forming $NH^+$ and $OH^-$ ions as $NH_3$ is basic.

$$NH_{3(aq)} + H_2O_{(l)} \rightleftharpoons NH_4^+{}_{(aq)} + OH^-{}_{(aq)}$$

Liquid $NH_3$ like $H_2O$ is a polarising solvent and it undergoes self-ionisation.

$$NH_{3(l)} + NH_{3(l)} \rightleftharpoons NH_4^+{}_{(l)} + NH_2^-{}_{(l)} \quad K_{NH_3} = 10^{-33}$$

$$H_2O_{(l)} + H_2O_{(l)} \rightleftharpoons H_3O^+{}_{(aq)} + OH^-{}_{(aq)} \quad K_W = 10^{-14}$$

Being basic ammonia reacts with acids forming ammonium salts.

$$NH_{3(aq)} + H^+{}_{(aq)} \longrightarrow NH_4^+{}_{(aq)}$$

The ammonium ion assumes a tetrahedral shape with bond angle of 109.5°C.

Its boiling point is higher than that of other group V hydrides due to the presence of strong intermolecular hydrogen bonds compared to weak Van der Waals intermolecular forces which operate between the other hydrides.

$NH_3$ also forms complex ions with several metal ions e.g

$$Ag^+{}_{(aq)} + 2NH_{3(aq)} \longrightarrow [Ag(NH_3)_2]^+{}_{(aq)}$$

$$Cu^{2+}{}_{(aq)} + 4NH_{3(aq)} \longrightarrow [Cu(NH_3)_4]^{2+}{}_{(aq)}$$

## Uses of $NH_3$

1. The major use of ammonia is manufacture of nitric acid from the catalytic oxidation of $NH_3$ using atmospheric oxygen and Pt-Rd catalyst at 900°C.

$$4NH_{3(g)} + 5O_{2(g)} \xrightleftharpoons[]{Pt-Rd\ catalyst,\ 7atm, 900^0C} 4NO_{(g)} + 6H_2O_{(g)} \quad \Delta H = -950KJmol^{-1}$$

$$2NO_{(g)} + O_{2(g)} \longrightarrow 2NO_{2(g)} \quad \Delta H = -114KJmol^{-1}$$

$$4NO_{2(g)} + 2H_2O_{(l)} + O_{2(g)} \longrightarrow 4HNO_{3(aq)}$$

2. The second major use of $NH_3$ is the manufacture of nitrogenous fertilizers by neutralisation with acids. The most common is ammonium nitrate formed by absorption of $NH_3$ in $HNO_3$.

$$NH_{3(g)} + HNO_{3(aq)} \longrightarrow NH_4NO_{3(aq)}$$

Ammonium nitrate when strongly heated decomposes explosively giving a mixture of $N_2$, $O_2$ and $H_2O$ – hence it's also used as an explosive.

$$NH_{4(s)} \xrightarrow{\Delta} N_{2(g)} + 2H_2O_{(g)} + \tfrac{1}{2}O_{2(g)}$$

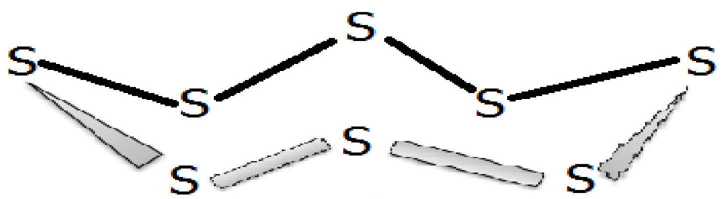
3. Manufacture of nylon and explosives e.g TNT and trinitrosylglycerine (nitroglycerine)
4. As a refrigerant.

**Problems with the over-use of fertilisers**

1. *Eutrophication:-* this is the abnormal enrichment of the waterways with mineral nutrients which encourage excessive uncontrolled growth of photosynthetic bacteria, algae and water plants. When these organisms die and decompose, they use oxygen from the water thus creating oxygen deficient in water leading to death of aquatic animals such as fish.
2. If the nitrates and nitrites gets into drinking water they oxidise the $Fe^{2+}$ of the haemoglobin to $Fe^{3+}$ reducing the ability of the haemoglobin to absorb oxygen in baby's blood a disease called methaemoglobianaemia-(blue-baby syndrome).

# SULPHUR

It's a yellow crystalline solid of low density which is insoluble in water with a melting point of +116°C. It exists as $S_8$ in its molecules in which S atoms are bonded covalently forming a puckered ring molecule.

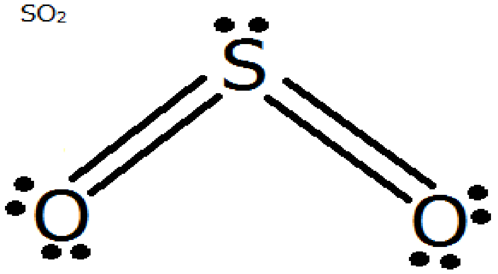


It's dimorphic that is it exists in 2 allotropic form which are the rhombic (which is stable below 96°C) and monoclinic (which is stable between 96-119°C) crystals.

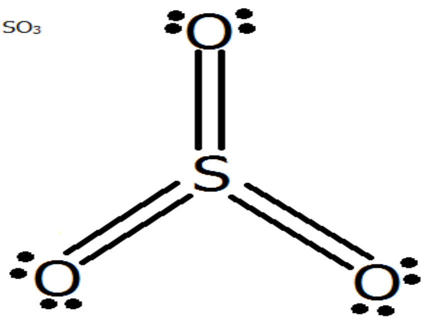
**Oxides of Sulphur**

Sulphur is a reactive element as it can be oxidised by elements like chlorine, fluorine and oxygen. It can also act as an oxidising agent when it reacts with elements of lower electronegativity such as iodine. Sulphur forms two main oxides $SO_2$ and $SO_3$ in which sulphur has oxidation state of +4 and +6 respectively.

$SO_2$



$SO_3$



$SO_2$ is a colourless gas with a pungent irritating odour, soluble in water forming a solution of sulphurous acid ($H_2SO_3$) which is readily decomposed so that all $SO_2$ can be recovered from solution by heating.

***Sulphur dioxide is a pollutant gas produced mainly from:-***

1. Roasting sulphur based ores such as chalcopyrites.

   $$CuFeS_{2(s)} + 3O_{2(g)} \longrightarrow FeO_{(s)} + CuO_{(s)} + SO_{2(g)}$$

2. Burning sulphur or fossil fuels containing sulphur in air such as coal or oil.

***It is a pollutant because it:***

1. Leads to the formation of acid rain which:-
   (a) is corrosive to buildings made of limestone, metal structures and destroys vegetation
   (b) Increases the acidity of water bodies thus killing aquatic life
   (c) Lowers the pH of agricultural soils reducing yield of crops
   (d) Increases the ion concentrations in water and soils

   $SO_2$ is oxidised in the atmosphere to give $SO_3$ which dissolves in rain giving $H_2SO_4$. The reaction is catalysed by another pollutant gas $NO_2$.

   $$SO_{2(g)} + NO_{2(g)} \longrightarrow SO_{3(g)} + NO_{(g)}$$
   $$NO_{(g)} + ½O_{2(g)} \longrightarrow NO_{2(g)}$$
   **Overall** $$SO_{2(g)} + ½O_{2(g)} \xrightarrow{NO_2} SO_{3(g)}$$
   $$SO_{3(g)} + H_2O_{(l)} \longrightarrow H_2SO_{4(aq)}$$
2. In areas where it is produced in high concentration it causes severe respiratory problems such as bronchitis, in which $SO_2$ dissolves in the moist respiratory tract forming $H_2SO_3$ which corrodes the respiratory surface.

Flue gases coming from these processes, containing $SO_2$ must be desulphurised before being released into the atmosphere. The desulphurisation is accomplished by passing the exhaust gases through an alkali which converts the $SO_2$ into waste sludge. MgO and CaO are often used in the desulphurisation.

$$MgO_{(s)} + SO_{2(g)} + ½O_{2(g)} \longrightarrow MgSO_{4(s)}$$
$$CaO_{(s)} + SO_{2(g)} + ½O_{2(g)} \longrightarrow CaSO_{4(s)}$$

The system removes up to 95% of the sulphur in the flue gases.

## Uses $SO_2$

$SO_2$ and $SO_3$ are used as food additives. They act as reducing agents to inhibit the oxidation of food components e.g unsaturated fats. $SO_2$ is oxidised to $SO_3$ and $SO_3^{2-}$ to $SO_4^{2-}$. They are also acidic and raise pH of food inhibiting bacteria and fungi growth.

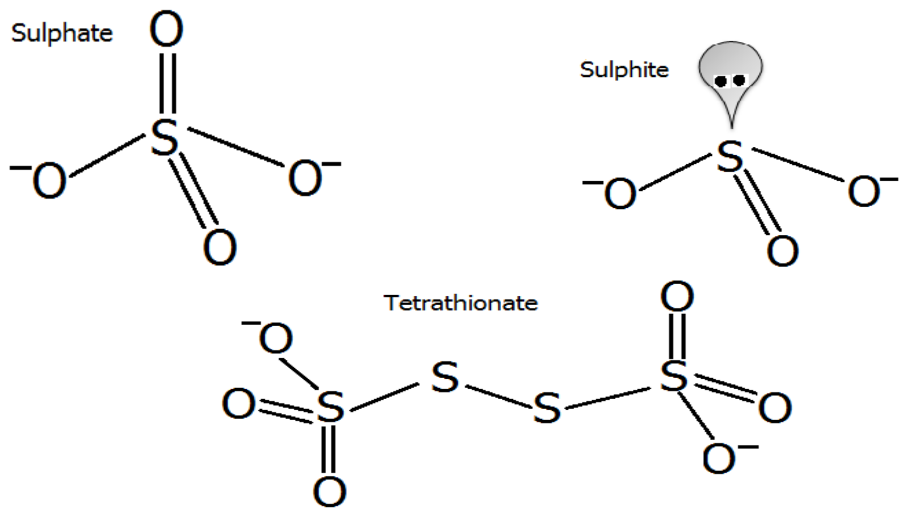
## Test for $SO_2$

$SO_2$ is tested by passing the gas through acidified $K_2Cr_2O_7$. It turns orange dichromate ions green as it reduces them to $Cr^{3+}$.

$$SO_2 + 2H_2O \rightleftharpoons SO_4^{2-} + 4H^+ + 2e^- \quad E^\theta = +0.17\ V$$
$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons Cr^{3+} + 7H_2O \quad E^\theta = +1.33\ V$$

**OVERALL** $$Cr_2O_7^{2-} + 2H^+ + 3SO_{2(g)} \longrightarrow 3SO_4^{2-} + 2Cr^{3+} + H_2O \quad E^\theta=+1.16\ V$$

S forms $SO_4^{2-}$ (sulphate), $SO_3^{2-}$ (sulphite), $S_2O_3^{2-}$ (thiosulphate), $S_4O_6^{2-}$ (tetrathionate), $S_2O_8^{2-}$ (peroxydisulphate )




Presence of $SO_4^{2-}$ and $SO_3^{2-}$ can be shown by adding $Ba^{2+}_{(aq)}$ ions e.g barium nitrate. They form white precipitates with barium ions.

$$Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \longrightarrow BaSO_{4(s)}$$

$$Ba^{2+}_{(aq)} + SO_3^{2-}{}_{(aq)} \longrightarrow BaSO_{3(s)}$$

The sulphate ions and sulphite ions can be distinguished from each other by adding dilute acids- sulphite precipitate dissolves in acid.

## Manufacture of Sulphuric acid by Contact Process

1. ***$SO_2$ production from burning pure sulphur or roasting sulphur based ores.***

$$S_{(s)} + O_{2(g)} \longrightarrow SO_{2(g)}$$

2. ***$SO_3$ production from $SO_2$***

$$2SO_{2(g)} + O_{2(g)} \xrightleftharpoons[450^0C,1-2\ atm]{Vanadium\ (V)oxide} SO_{3(g)} \quad \Delta H = -197KJmol^{-1}$$

The forward reaction being exothermic is favoured by low temperature but at very low temperature the rate will be too slow hence a moderate temperature is used. Very high pressures are avoided because at high pressure $SO_2$ liquifies and also at 450°C and 2 atm a high yield of $SO_3$ about 95% is obtained.

3. ***Oleum production by dissolving $SO_3$ in concentrated $H_2SO_4$***

$$H_2SO_{4(conc)} + SO_{3(g)} \longrightarrow \underset{(oleum)}{H_2S_2O_{7(l)}}$$

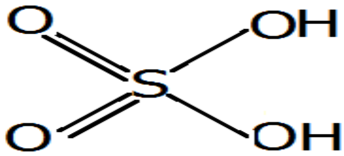
4. ***Production of conc. $H_2SO_4$ (98%) by dilution of $H_2S_2O_7$ with $H_2O$***

$$H_2S_2O_{7(l)} + H_2O_{(l)} \longrightarrow 2H_2SO_{4(l)}$$

***NB:-*** $SO_3$ is not reacted directly with water as the reaction is very vigorous and produces a mist of $H_2SO_4$ which is difficult to condense.

**Properties of sulphuric acid**

Pure sulphuric acid is an oily liquid with high density; it forms colourless crystals below 10°C. It is a covalent substance with the following structure.



The molecule is tetrahedral in shape with extensive H-bonding between the molecules both in solid and liquid state hence its high viscosity and density.

In aqueous solutions it ionises giving hydrogen ions. The reaction is highly exothermic, that is why sulphuric acid is always added to water during dilution and not water to acid.

$$H_2SO_4 \rightleftharpoons H^+ + HSO_4^-$$

$$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$$

Concentrated sulphuric acid is a fairly strong oxidising agent and also a dehydrating agent.

**Uses of sulphuric acid**

1. Manufacture of fertilisers super phosphates
2. Manufacture of paints and dyes
3. Manufacture of detergents
4. Manufacture of plastics and fibres
5. Manufacture of explosives and for metal processing